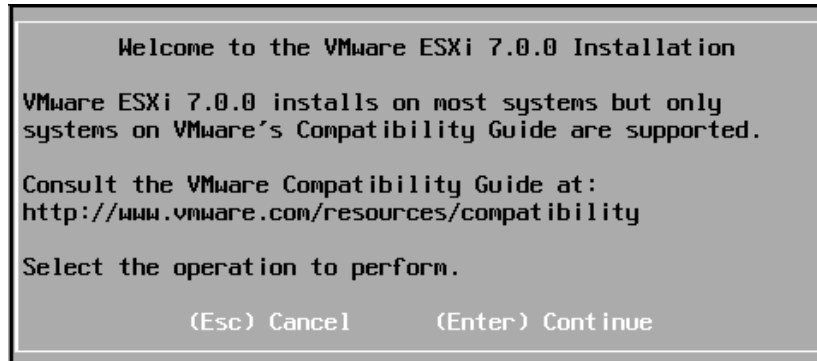


VMware ESXi 7.0

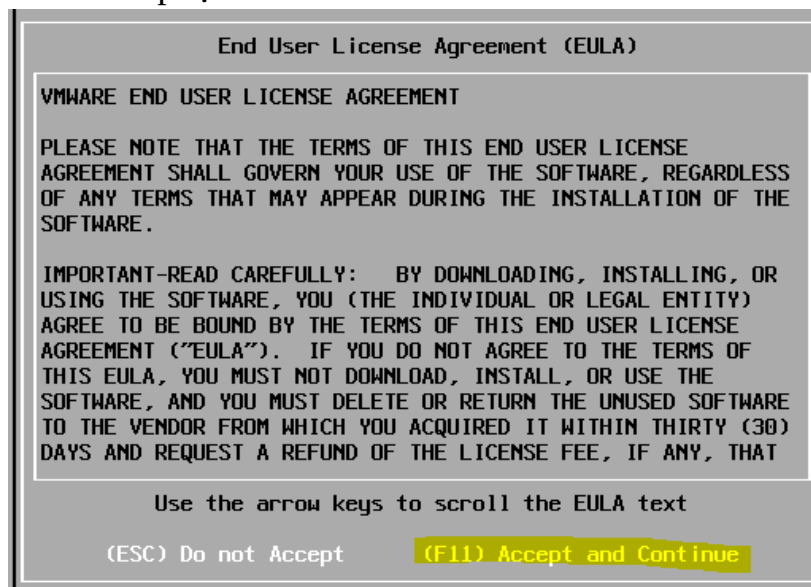
1. Hướng dẫn cài đặt

- Chỉnh boot từ file Iso hoặc ghi ra USB

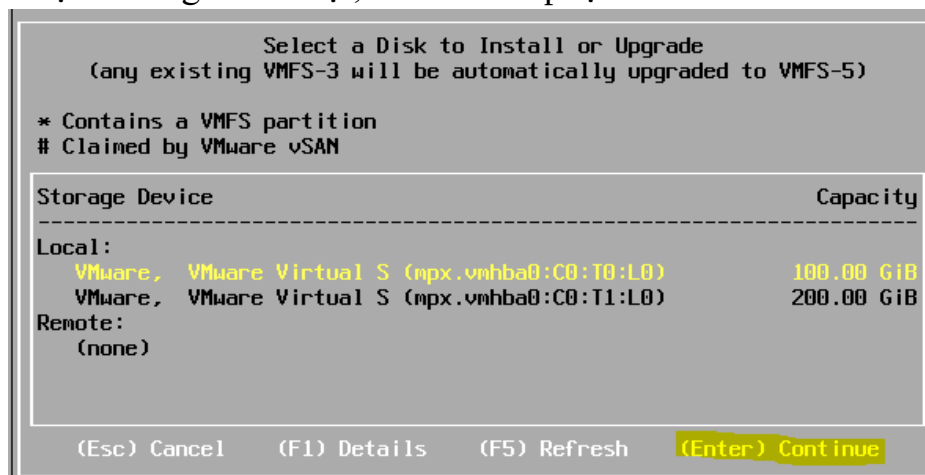
Enter để tiếp tục



F11 để tiếp tục



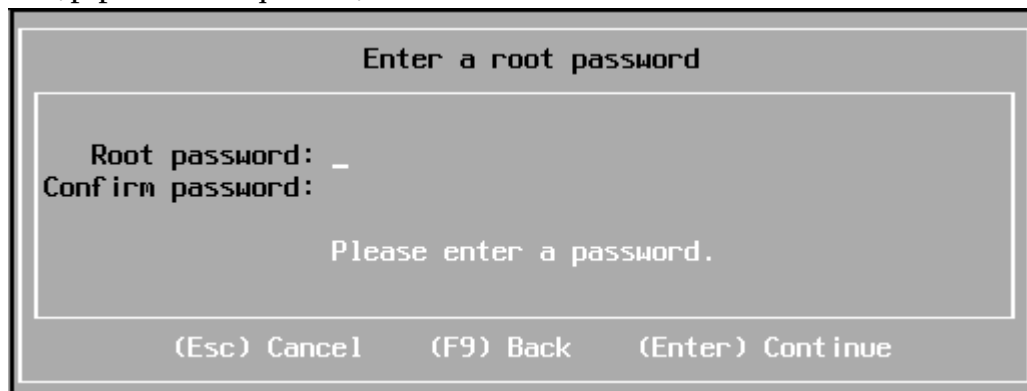
Chọn ổ cứng để cài đặt, Enter để tiếp tục



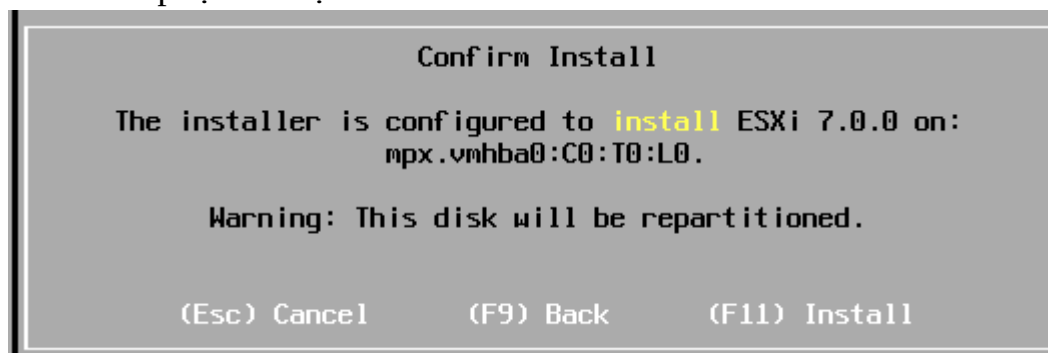
Enter để tiếp tục



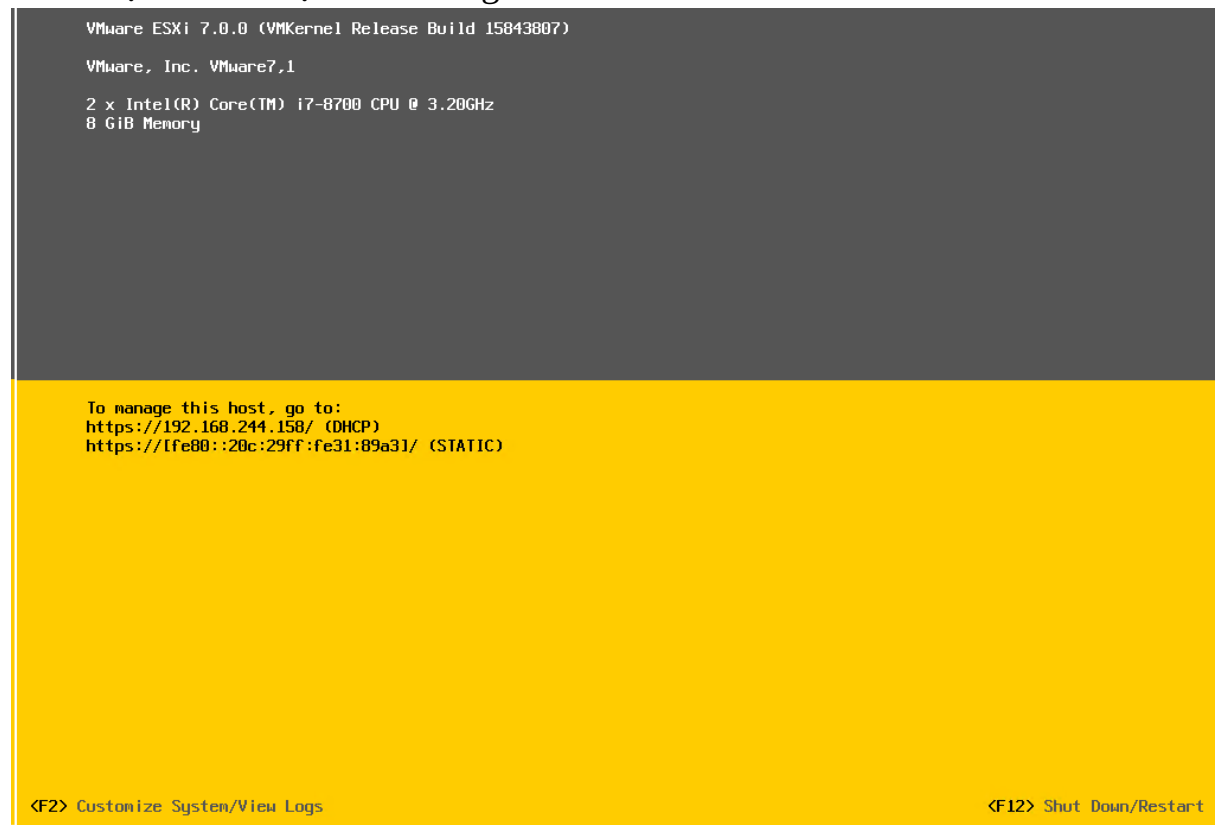
Nhập password quản trị



F11 để tiếp tục cài đặt



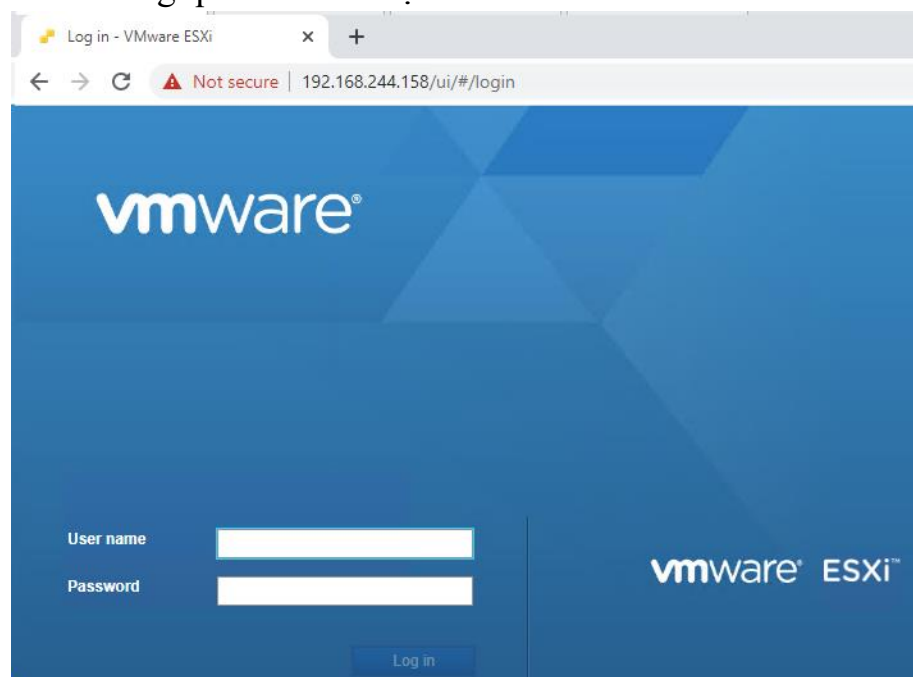
Giao diện khi cài đặt thành công



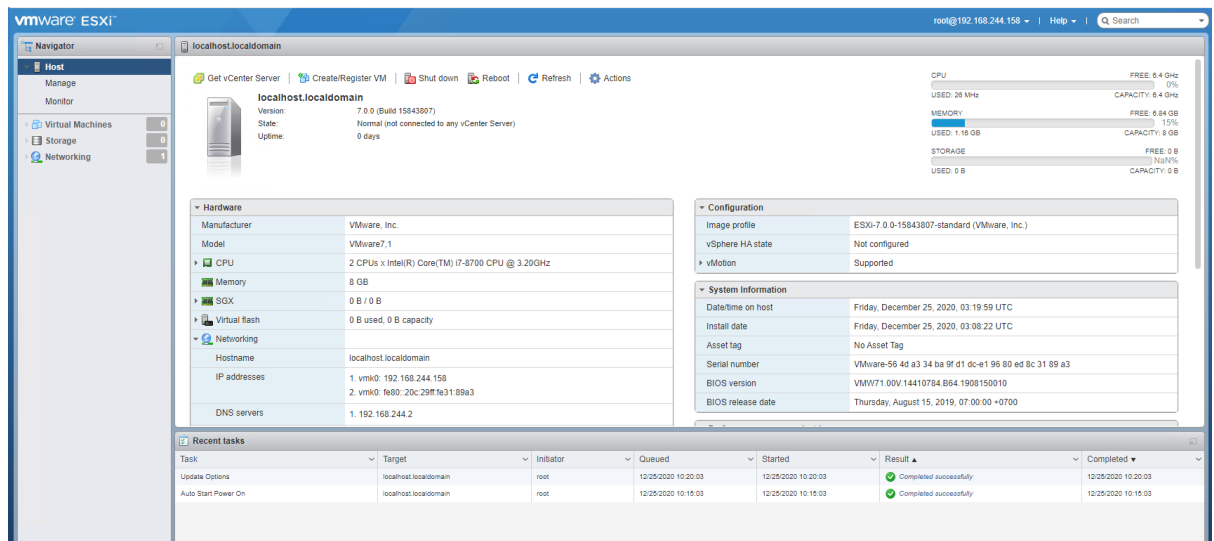
Dùng 1 máy windows mở trình duyệt web để vào trang quản trị

User: root

Pass: trong quá trình cài đặt

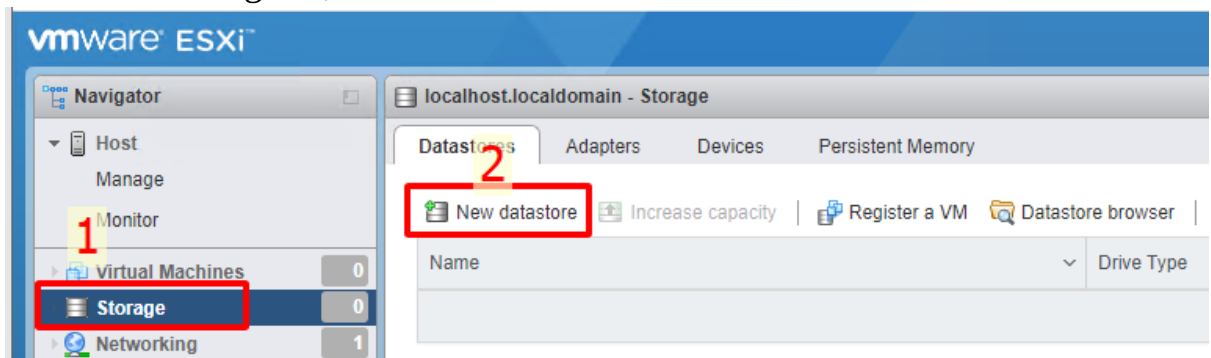


Giao diện chính của vmware ESXi 7.0

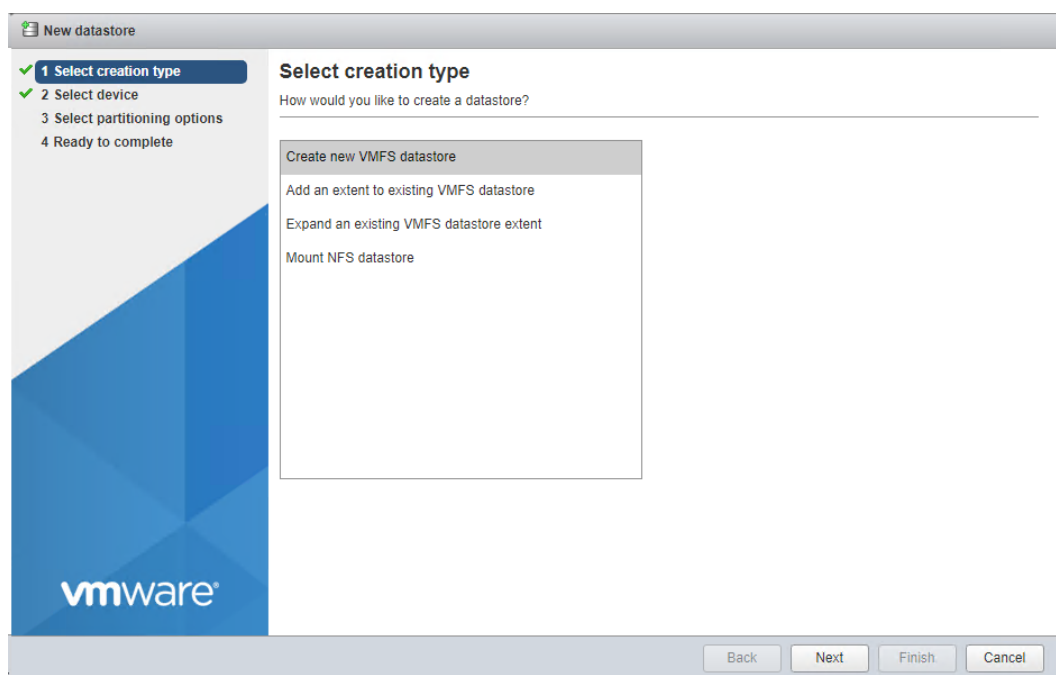


2. Hướng dẫn cài VM

- Trước tiên chúng ta tạo thêm 1 datastore để chứa VM và source



Next để tiếp tục



Đặt tên storage và chọn ổ cứng

New datastore - Datastore2

1 Select creation type
2 Select device
3 Select partitioning options
4 Ready to complete

Select device

Select a device on which to create a new VMFS partition

Name
Datastore2

The following devices are unclaimed and can be used to create a new VMFS datastore

Name	Type	Capacity	Free space
Local VMware, Disk (mpx.vmhba0:C0:T1:L0)	Disk	200 GB	200 GB

1 items

Chọn dung lượng muốn sử dụng ở đây mình chọn full

New datastore - datastore2

1 Select creation type
2 Select device
3 Select partitioning options
4 Ready to complete

Select partitioning options

Select how you would like to partition the device

Use full disk VMFS 6

Before, select a partition

Free space (200 GB)

After

1. VMFS (200 GB)

Back Next Finish Cancel

New datastore - datastore2

1 Select creation type
2 Select device
3 Select partitioning options
4 Ready to complete

Ready to complete

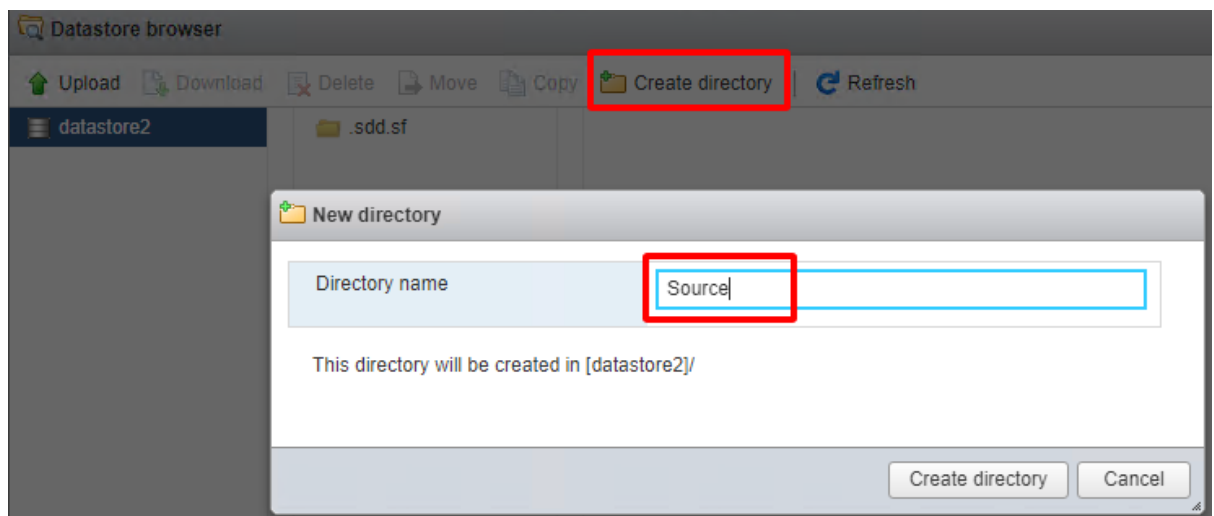
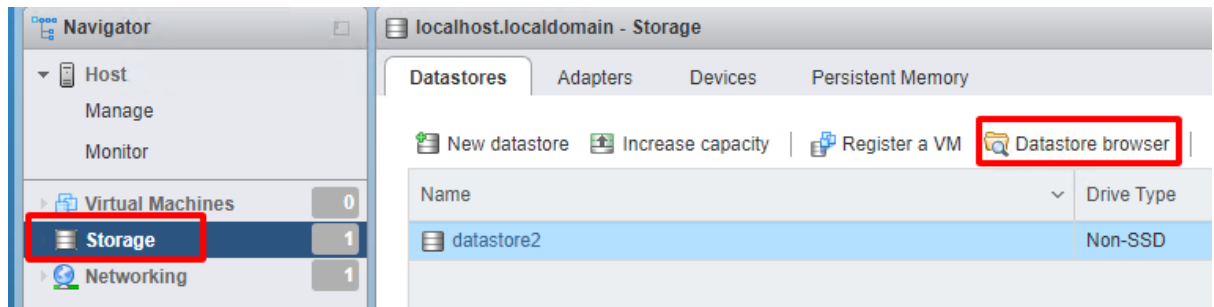
Summary

Name	datastore2
Disk	Local VMware, Disk (mpx.vmhba0:C0:T1:L0)
Partitioning	Use full disk
VMFS version	6

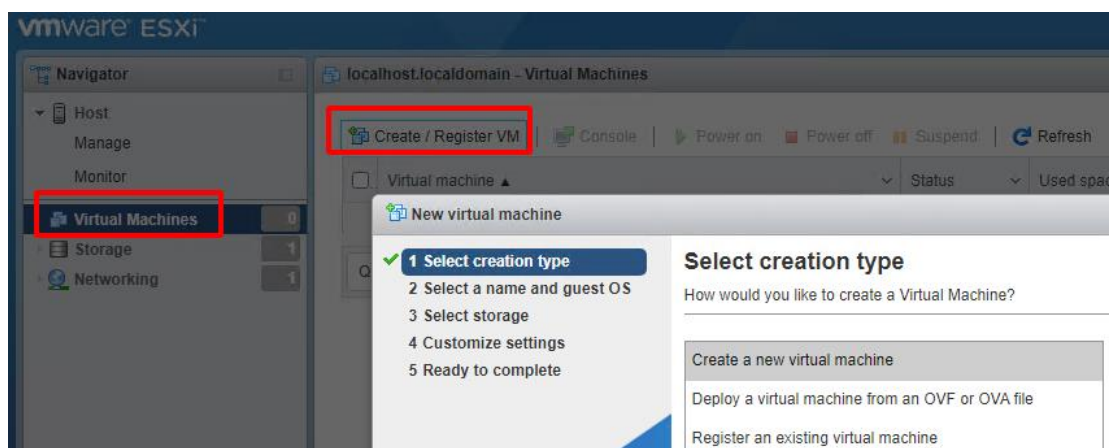
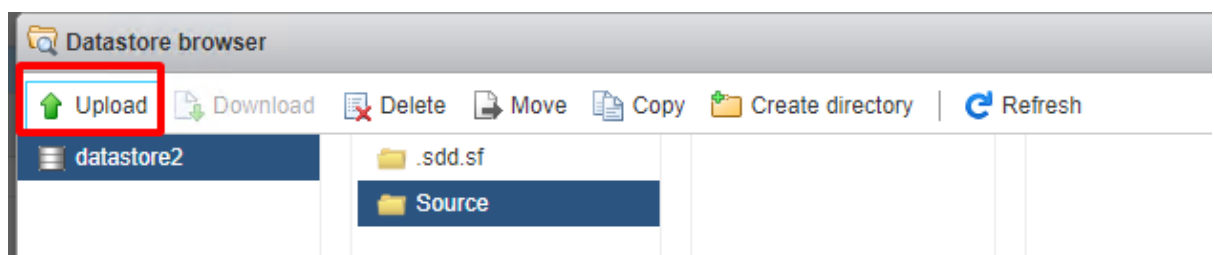
VMFS (200 GB)

Back Next Finish Cancel

- Tạo thư mục source trên datastore2 để chứa file cài đặt



Chọn upload và tìm file Iso của hệ điều hành muốn cài đặt



Chọn các thông tin phù hợp với hệ điều hành cần cài

New virtual machine - Windows_server_2019 (ESXi 7.0 virtual machine)

1 Select creation type
2 Select a name and guest OS
3 Select storage
4 Customize settings
5 Ready to complete

Select a name and guest OS

Specify a unique name and OS

Name
Windows_server_2019

Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.

Identifying the guest operating system here allows the wizard to provide the appropriate defaults for the operating system installation.

Compatibility
ESXi 7.0 virtual machine

Guest OS family
Windows

Guest OS version
Microsoft Windows Server 2019 (64-bit)

☐ Enable Windows Virtualization Based Security ⓘ

Chọn datastore muốn cài đặt

New virtual machine - Windows_server_2019 (ESXi 7.0 virtual machine)

1 Select creation type
2 Select a name and guest OS
3 Select storage
4 Customize settings
5 Ready to complete

Select storage

Select the storage type and datastore

Standard Persistent Memory

Select a datastore for the virtual machine's configuration files and all of its' virtual disks.

Name	Capacity	Free	Type	Thin pro...	Access
datastore2	199.75 GB	198.34 GB	VMFS6	Supported	Single

1 items

Chọn cấu hình phù hợp với yêu cầu

New virtual machine - Windows_server_2019 (ESXi 7.0 virtual machine)

1 Select creation type
2 Select a name and guest OS
3 Select storage
4 Customize settings
5 Ready to complete

Customize settings

Configure the virtual machine hardware and virtual machine additional options

Virtual Hardware VM Options

CPU 2 ⓘ

Memory 4096 MB

Hard disk 1 90 GB

SCSI Controller 0 LSI Logic SAS

SATA Controller 0

USB controller 1 USB 3.1

Network Adapter 1 VM Network ☒ Connect

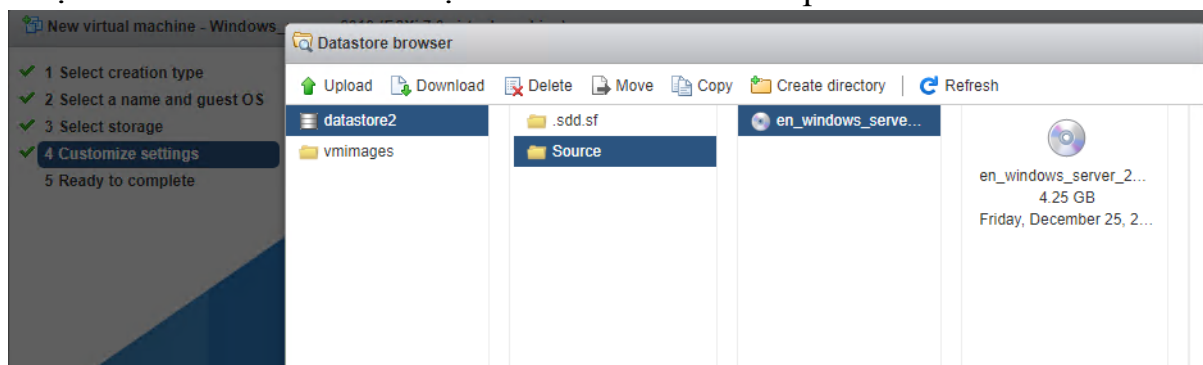
CD/DVD Drive 1 Host device ☒ Connect

Video Card

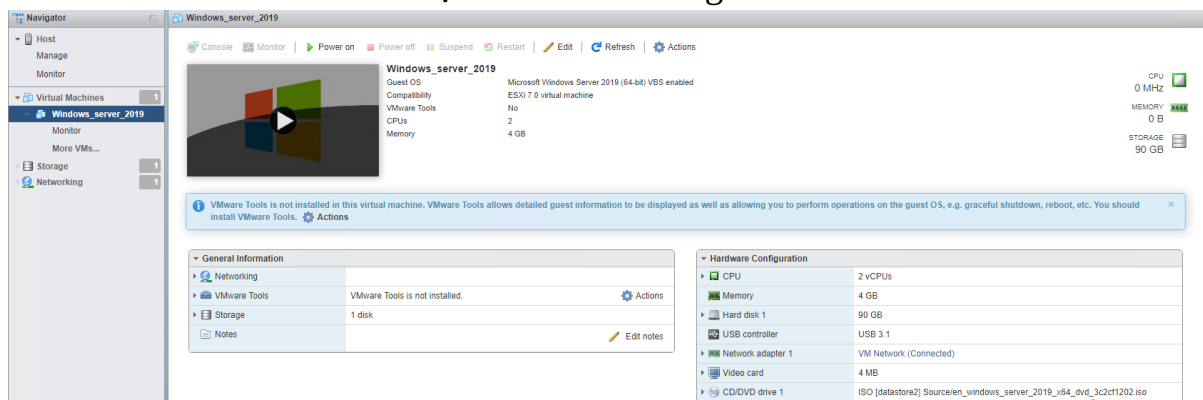
Host device
Datastore ISO file

Back Next Finish Cancel

Chọn datastore ISO file > chọn nơi chứa file iso đã upload ở trên



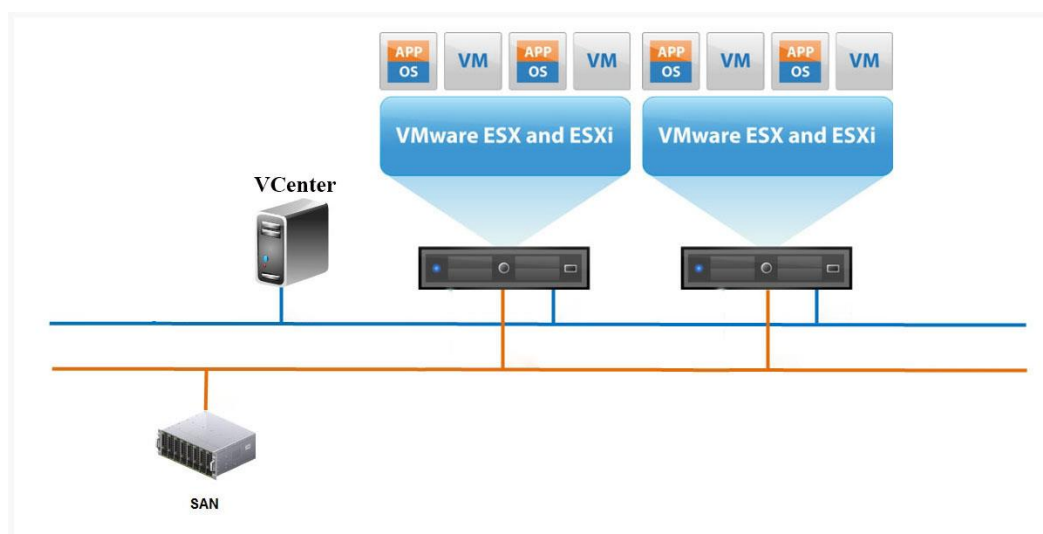
Tiến hành start VM lên cài đặt như bình thường



3. Hướng dẫn HA trên VMware ESXi

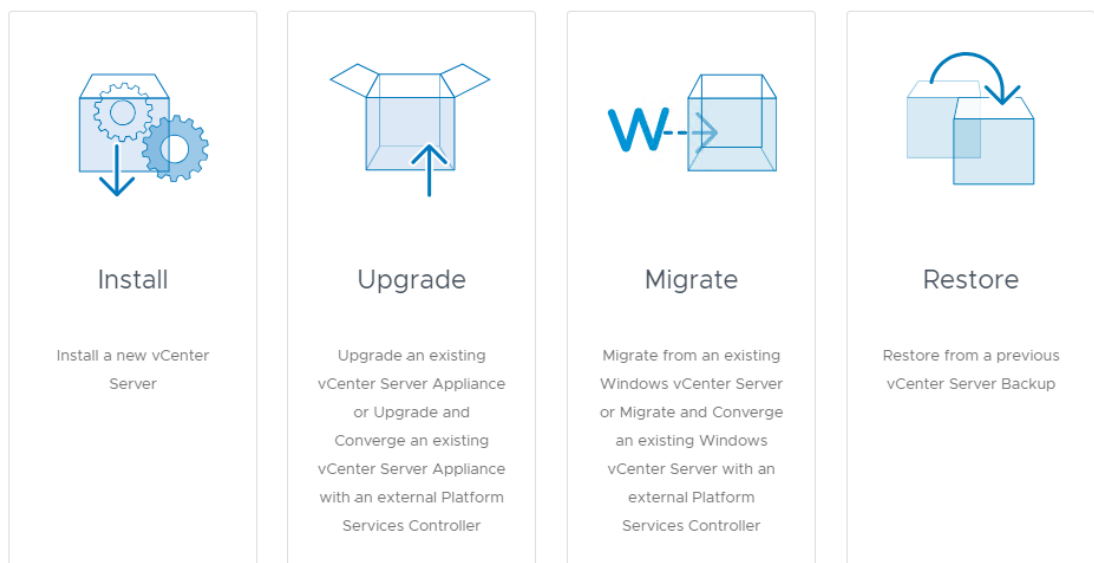
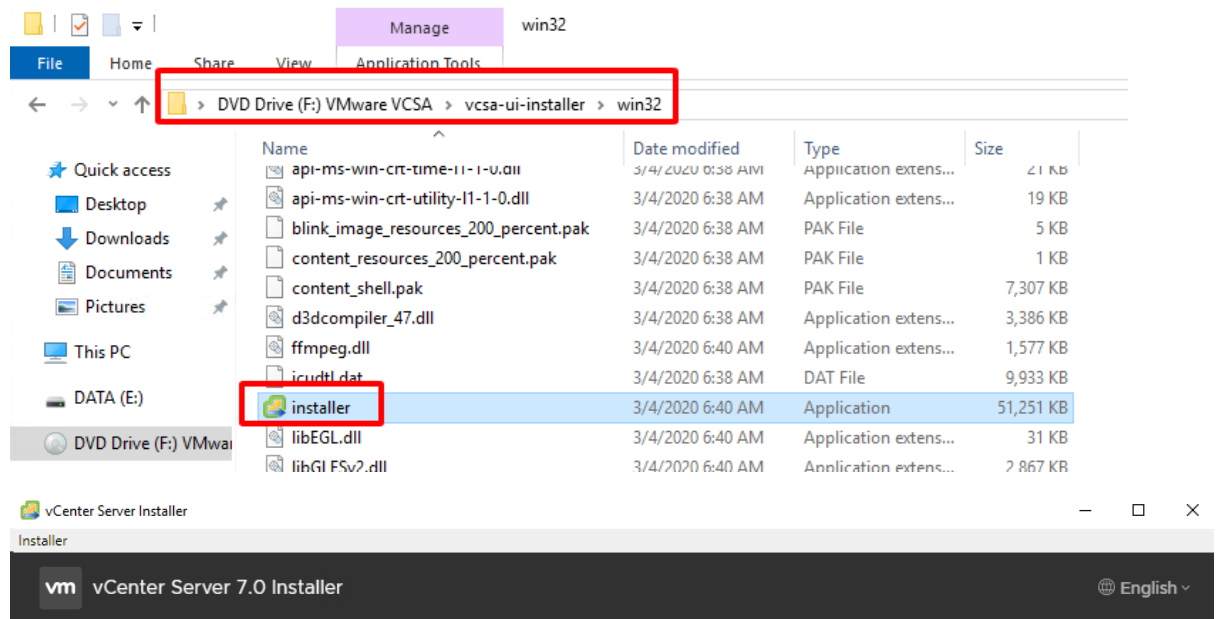
Để sử dụng được HA thì phải có:

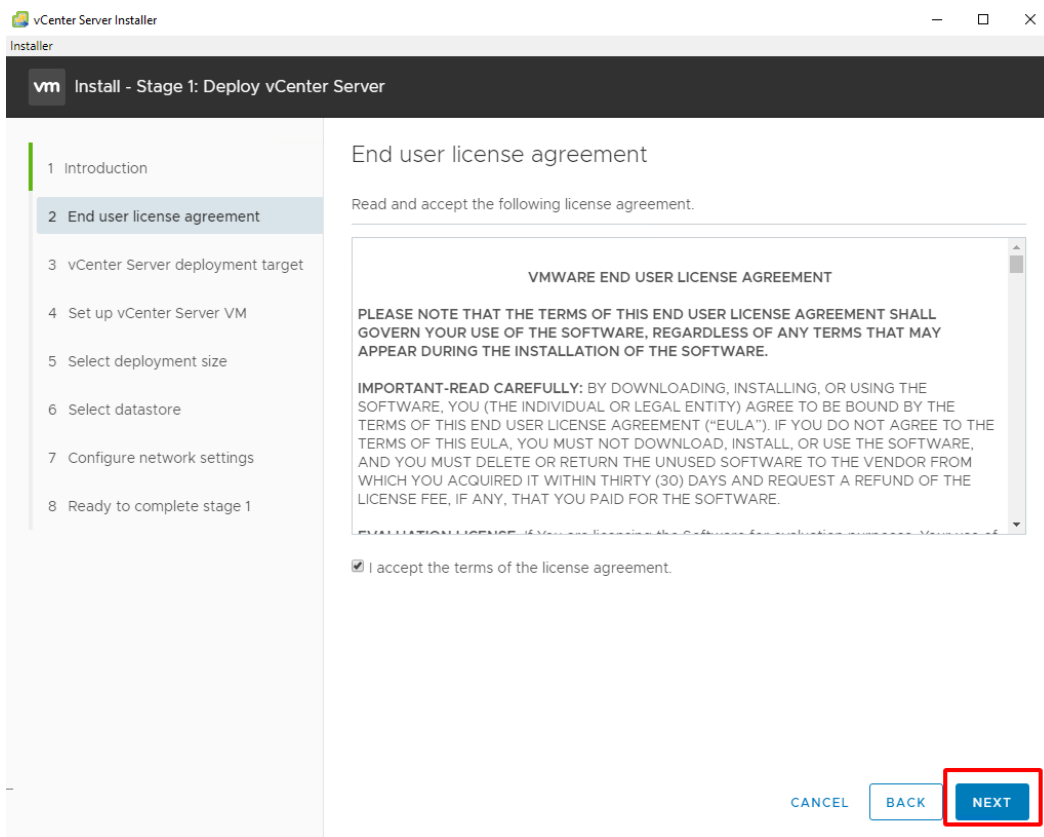
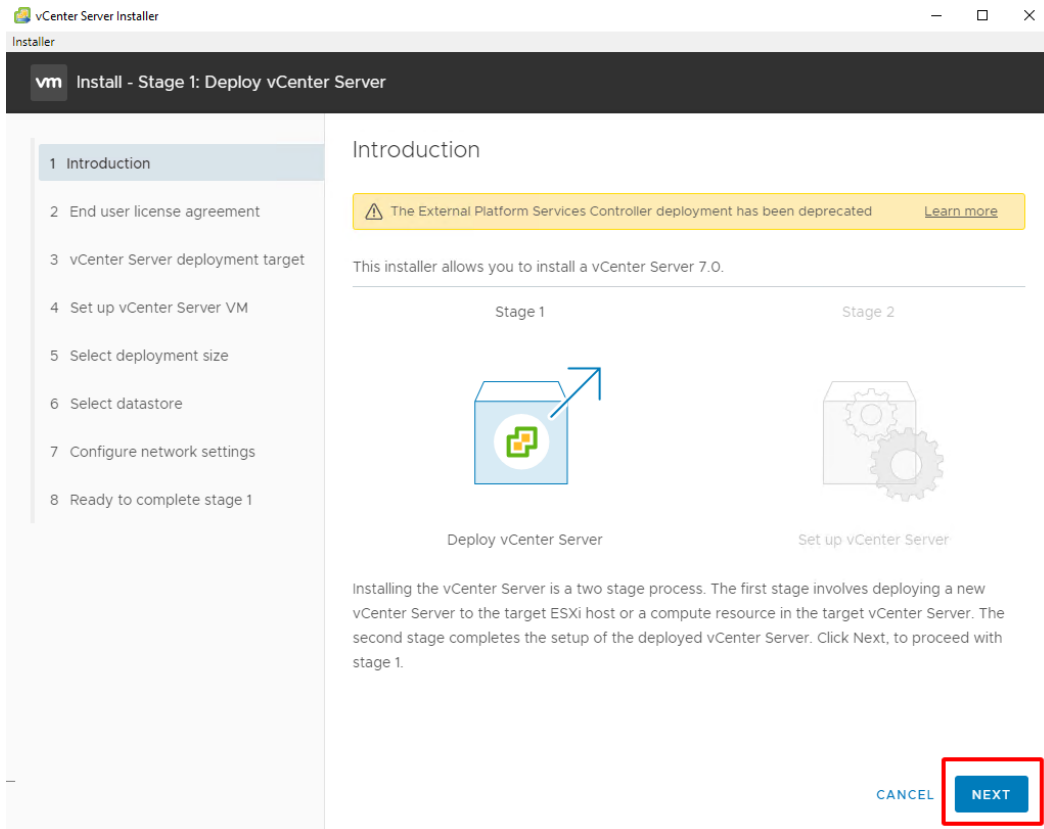
- Storage lưu trữ tập trung ở bài này mình sẽ dùng FreeNas để giả lập SAN chạy iSCSI.
- VCenter để quản lý tập trung các Node
- Mô hình bài lab



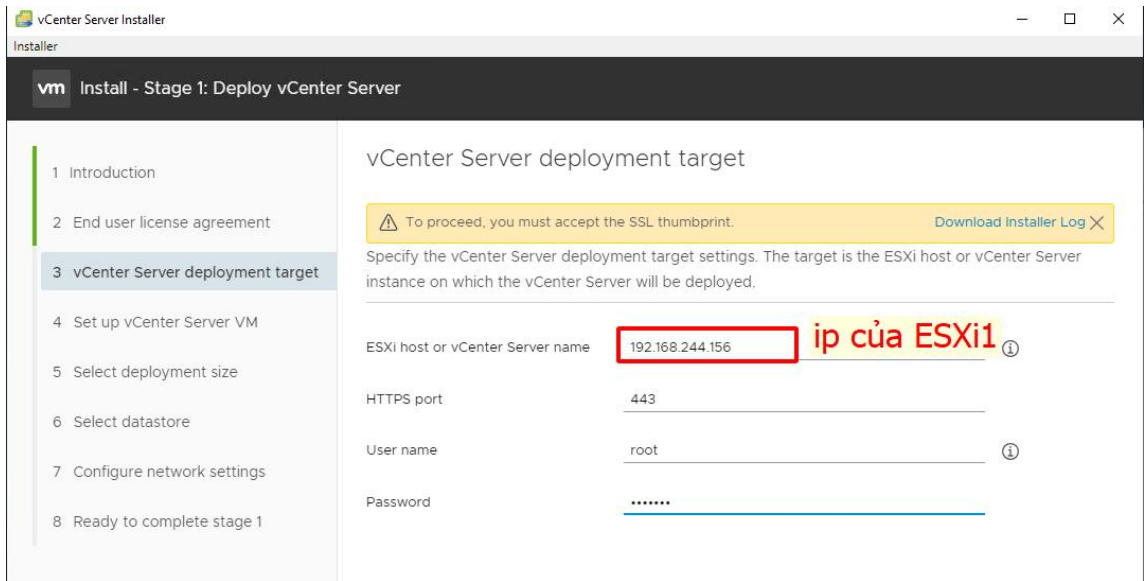
- 3.1 Hướng dẫn cài VCenter trên windows

Tải file cài đặt từ trang chủ, Mount ra ổ ảo rồi tiến hành cài đặt





Điền thông tin IP, tài khoản và password của ESXi1



The screenshot shows the 'vCenter Server deployment target' screen. A sidebar on the left lists steps 1 through 8, with step 3 'vCenter Server deployment target' highlighted. The main area contains a warning about SSL thumbprint, instructions to specify deployment target settings, and input fields for ESXi host name, HTTPS port, user name, and password. The ESXi host name '192.168.244.156' is highlighted with a red box and labeled 'ip của ESXi1'.

vCenter Server deployment target

To proceed, you must accept the SSL thumbprint. [Download Installer Log](#)

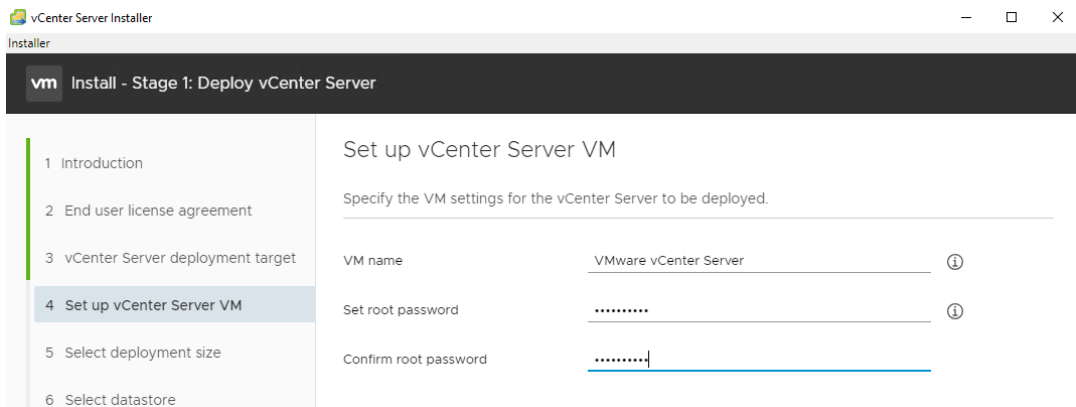
Specify the vCenter Server deployment target settings. The target is the ESXi host or vCenter Server instance on which the vCenter Server will be deployed.

ESXi host or vCenter Server name: 192.168.244.156 ip của ESXi1

HTTPS port: 443

User name: root

Password:



The screenshot shows the 'Set up vCenter Server VM' screen. The sidebar highlights step 4 'Set up vCenter Server VM'. The main area contains instructions to specify VM settings and input fields for VM name, root password, and confirm root password.

Set up vCenter Server VM

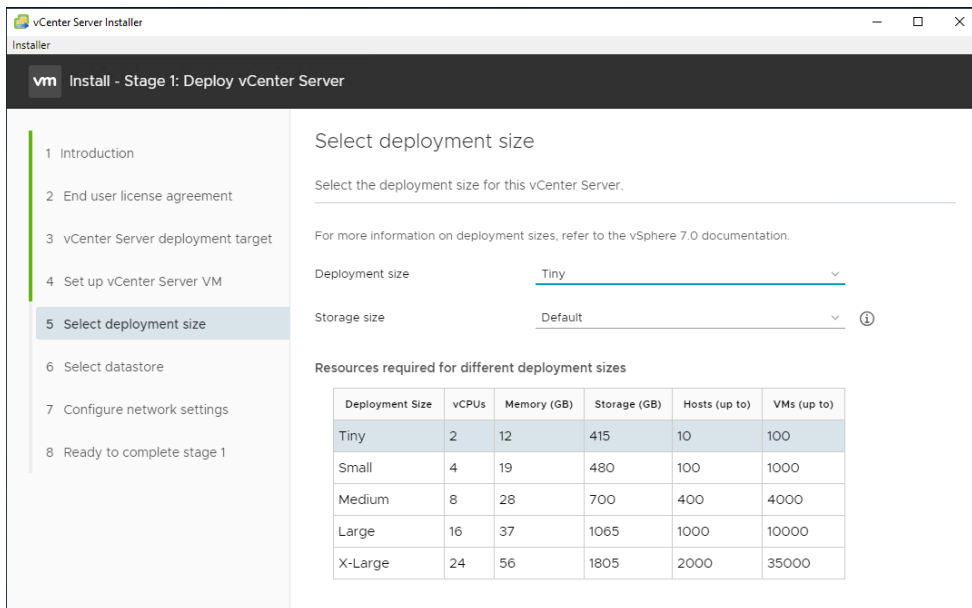
Specify the VM settings for the vCenter Server to be deployed.

VM name: VMware vCenter Server

Set root password:

Confirm root password:

Chọn thông tin cấu hình phù hợp



The screenshot shows the 'Select deployment size' screen. The sidebar highlights step 5 'Select deployment size'. The main area contains instructions to select deployment size, a table of resources required for different deployment sizes, and dropdown menus for deployment and storage size.

Select deployment size

Select the deployment size for this vCenter Server.

For more information on deployment sizes, refer to the vSphere 7.0 documentation.

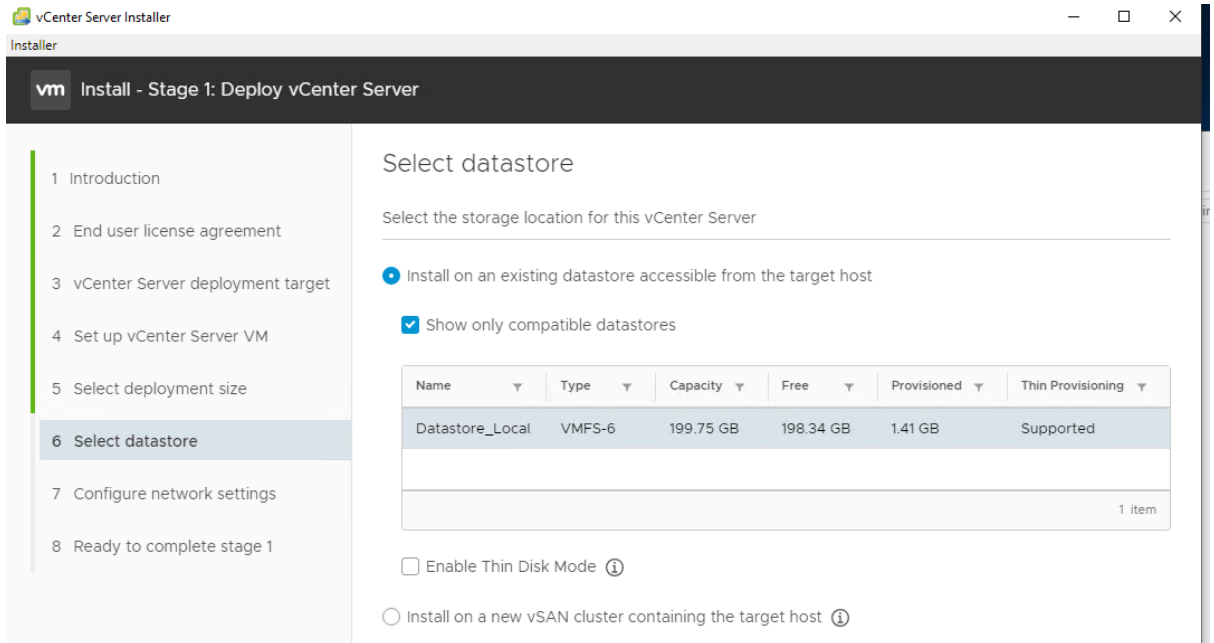
Deployment size: Tiny

Storage size: Default

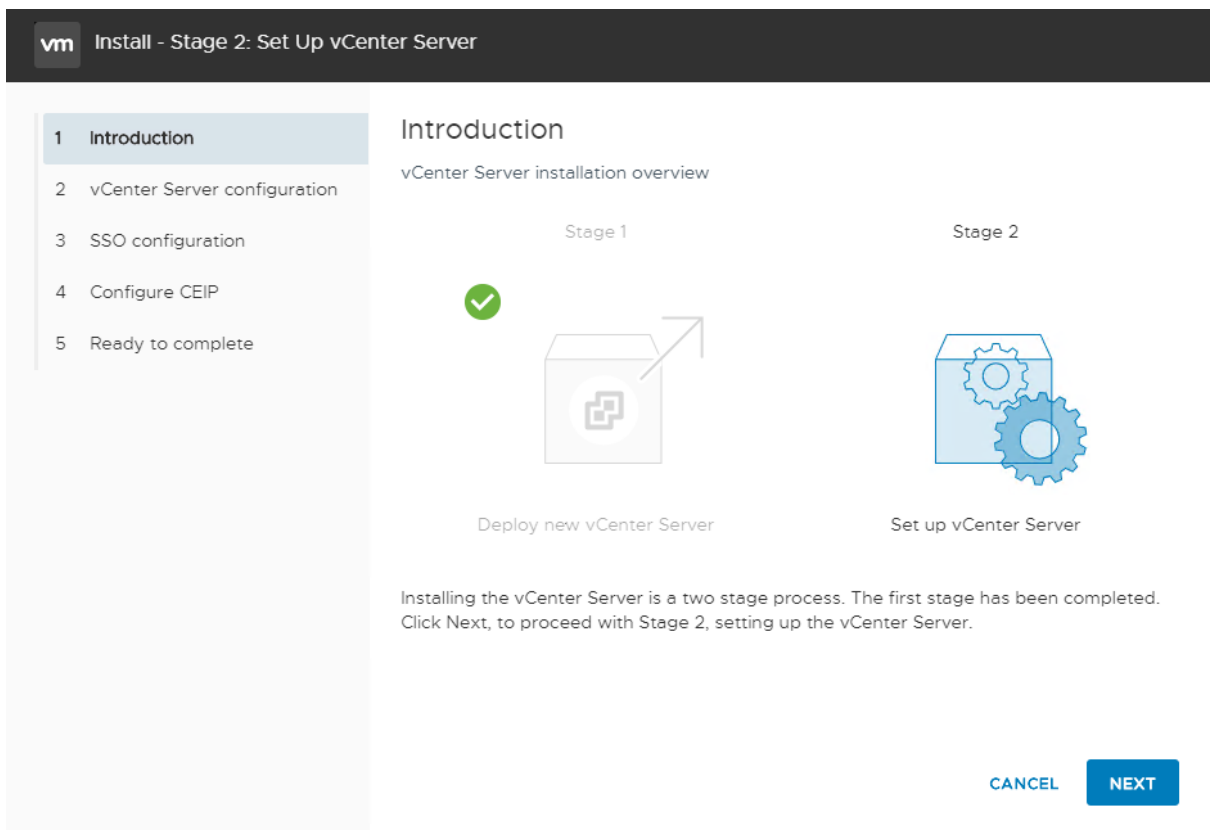
Resources required for different deployment sizes

Deployment Size	vCPUs	Memory (GB)	Storage (GB)	Hosts (up to)	VMs (up to)
Tiny	2	12	415	10	100
Small	4	19	480	100	1000
Medium	8	28	700	400	4000
Large	16	37	1065	1000	10000
X-Large	24	56	1805	2000	35000

Chọn datastore để tiến hành deploy



Giai đoạn 2 – Cấu hình dịch vụ VCSA 7.0



vm

Install - Stage 2: Set Up vCenter Server

1

Introduction

2

vCenter Server configuration

3

SSO configuration

4

Configure CEIP

5

Ready to complete

vCenter Server configuration

Time synchronization mode

Synchronize time with the ESXi ho

SSH access

Enabled

CANCEL

BACK

NEXT

Nhập thông tin SSO để quản lý đăng nhập vào vCenter Web Client để quản lý hạ tầng vSphere máy chủ ảo.

vm

Install - Stage 2: Set Up vCenter Server

1

Introduction

2

vCenter Server configuration

3

SSO configuration

4

Configure CEIP

5

Ready to complete

SSO configuration

Create a new SSO domain

Single Sign-On domain name

vccenter.local

Single Sign-On user name

administrator

Single Sign-On password

Confirm password

Join an existing SSO domain

PSC

vCenter

CANCEL

BACK

NEXT

- 1 Introduction
- 2 vCenter Server configuration
- 3 SSO configuration
- 4 Configure CEIP**
- 5 Ready to complete

Configure CEIP

Join the VMware Customer Experience Improvement Program

Participating in VMware's Customer Experience Improvement Program ("CEIP") enables VMware to provide you with a proactive, reliable, and consistent vSphere environment and experience. Examples of such enhancements can be seen in the following features:

- vSphere Health
- vSAN Online Health
- vCenter Server Update Planner
- vSAN Performance Analytics
- Host Hardware Compatibility
- vSAN Support Insight

CEIP collects configuration, feature usage, and performance information. No personally identifiable information is collected. All data is sanitized and obfuscated prior to being received by VMware.

For additional information on CEIP and the data collected, please see VMware's

☒ Join the VMware's Customer Experience Improvement Program (CEIP)

CANCEL

BACK

NEXT

- 1 Introduction
- 2 vCenter Server configuration
- 3 SSO configuration
- 4 Configure CEIP
- 5 Ready to complete**

Ready to complete

Review your settings before finishing the wizard.

Network Details

Network configuration	Assign static IP address
IP version	IPv4
Host name	localhost
IP Address	192.168.244.100
Subnet mask	255.255.255.0
Gateway	192.168.244.2
DNS servers	192.168.244.2

vCenter Server Details

Time synchronization mode	Synchronize time with the ESXi host
SSH access	Enabled

SSO Details

Domain name	vcenter.local
User name	administrator

Customer Experience Improvement Program

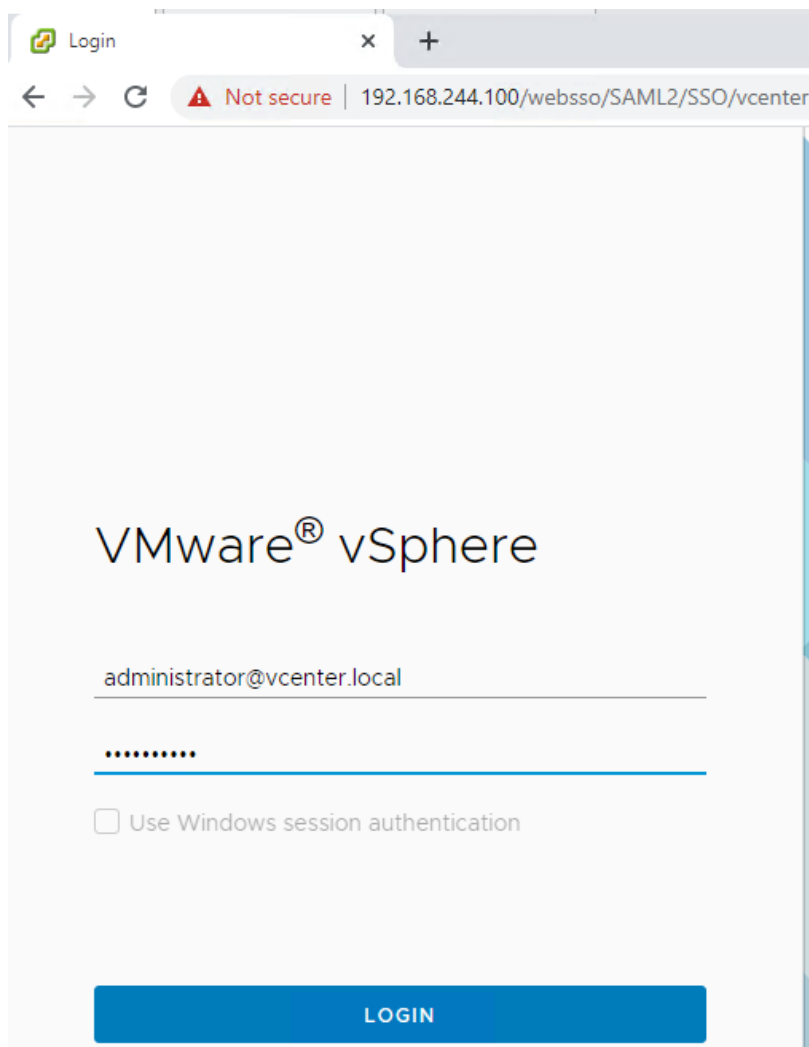
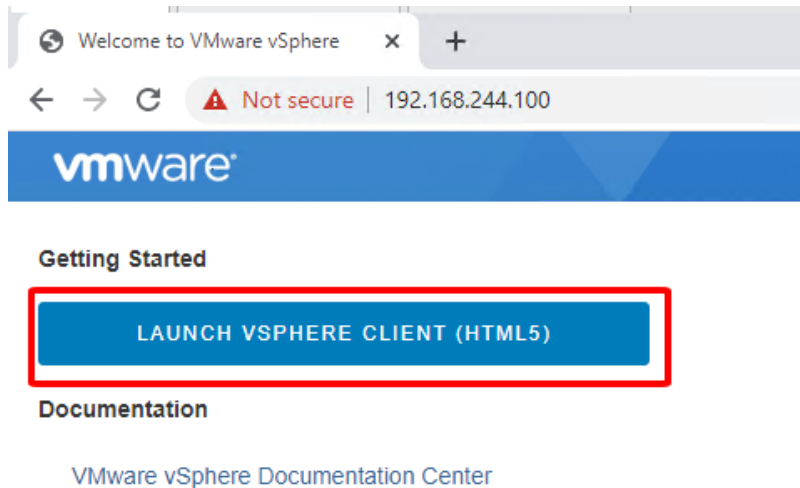
CANCEL

BACK

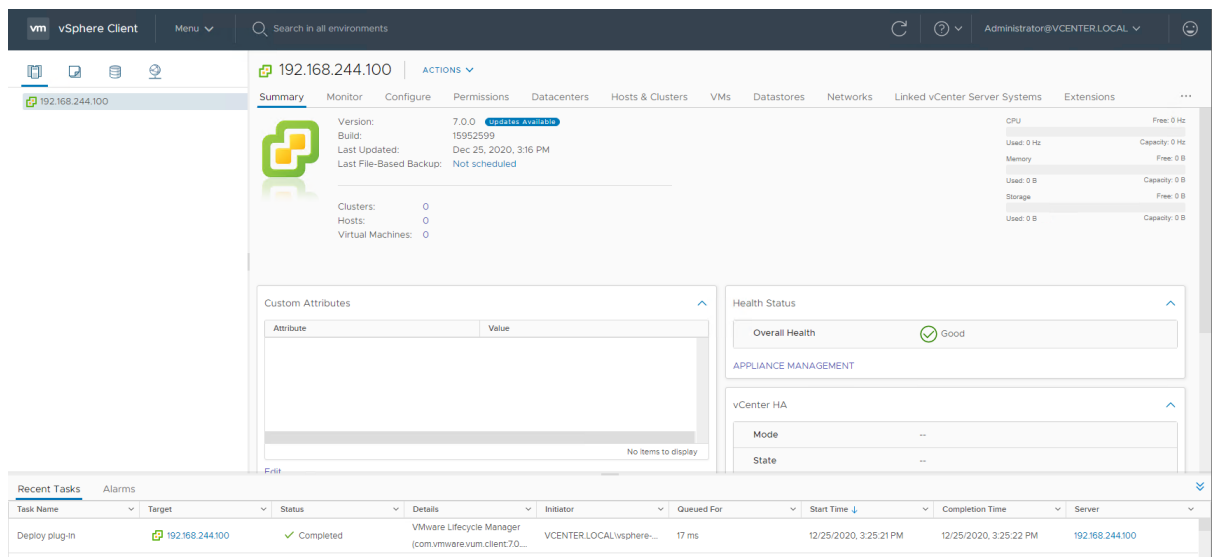
FINISH

Sau khi cài đặt hoàn tất ta tiến hành mở trình duyệt web đăng nhập vào trang quản trị

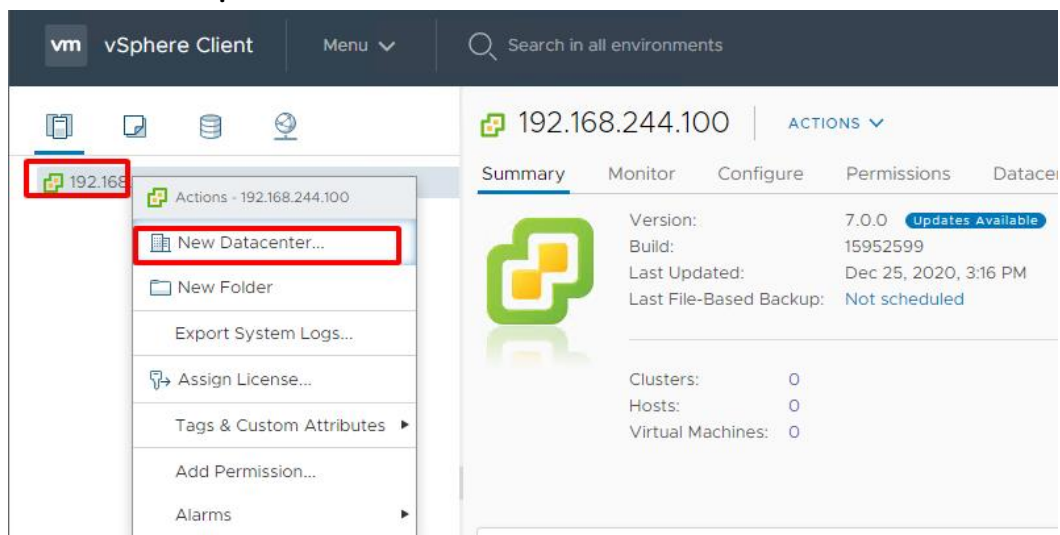
<https://192.168.244.100/>



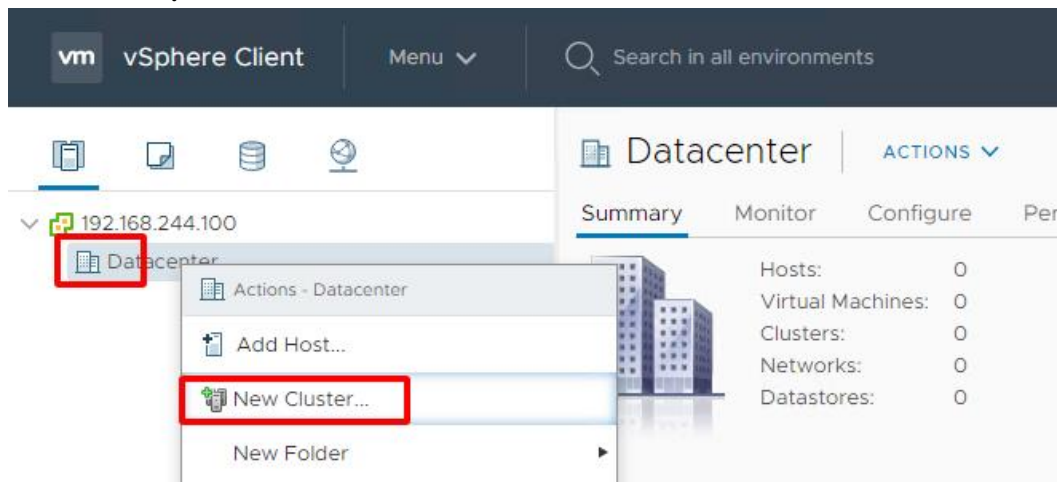
Giao diện chính của Vcenter



- Ta tiến hành tạo New Datacenter



- Sau đó ta tạo New Cluster



New Cluster | Datacenter

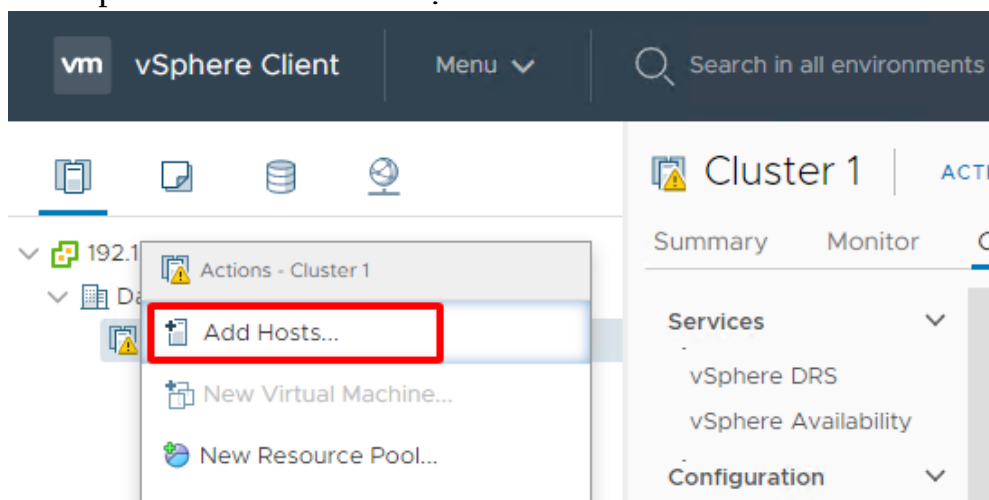
Name	New Cluster
Location	Datacenter
vSphere DRS	☒
vSphere HA	☒
vSAN	☒

These services will have default settings - these can be changed later in the Cluster Quickstart workflow.

☐ Manage all hosts in the cluster with a single image

CANCEL OK

- Click phải vào Cluster vừa tạo > Add Hosts..



Tiến hành điền thông tin của từng ESXi muốn thêm vào Cluster

Add hosts

Add new and existing hosts to your cluster

1 Add hosts

2 Host summary

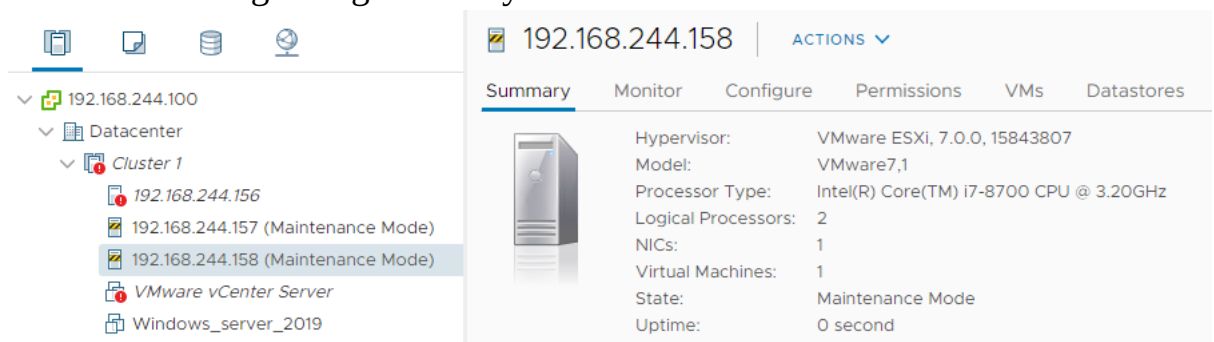
3 Ready to complete

New hosts (2) Existing hosts (0 from 0)

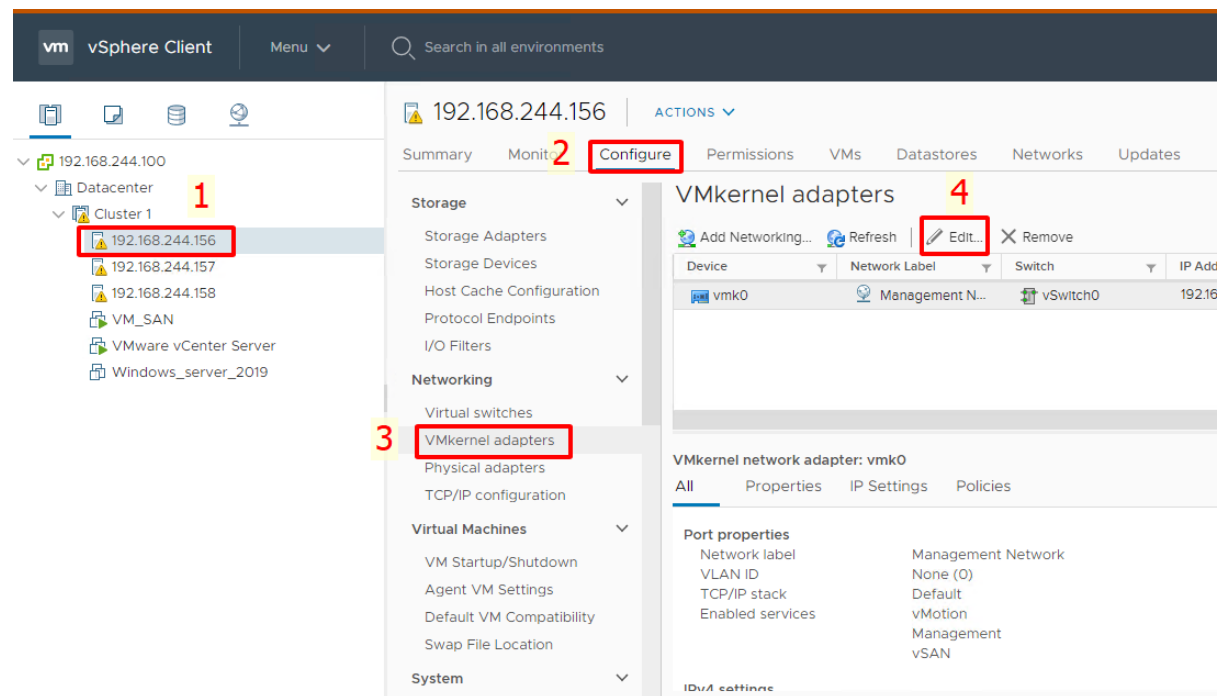
☐ Use the same credentials for all hosts

192.168.244.156	root		x
192.168.244.157	root		x
IP address or FQDN	Username	Password	

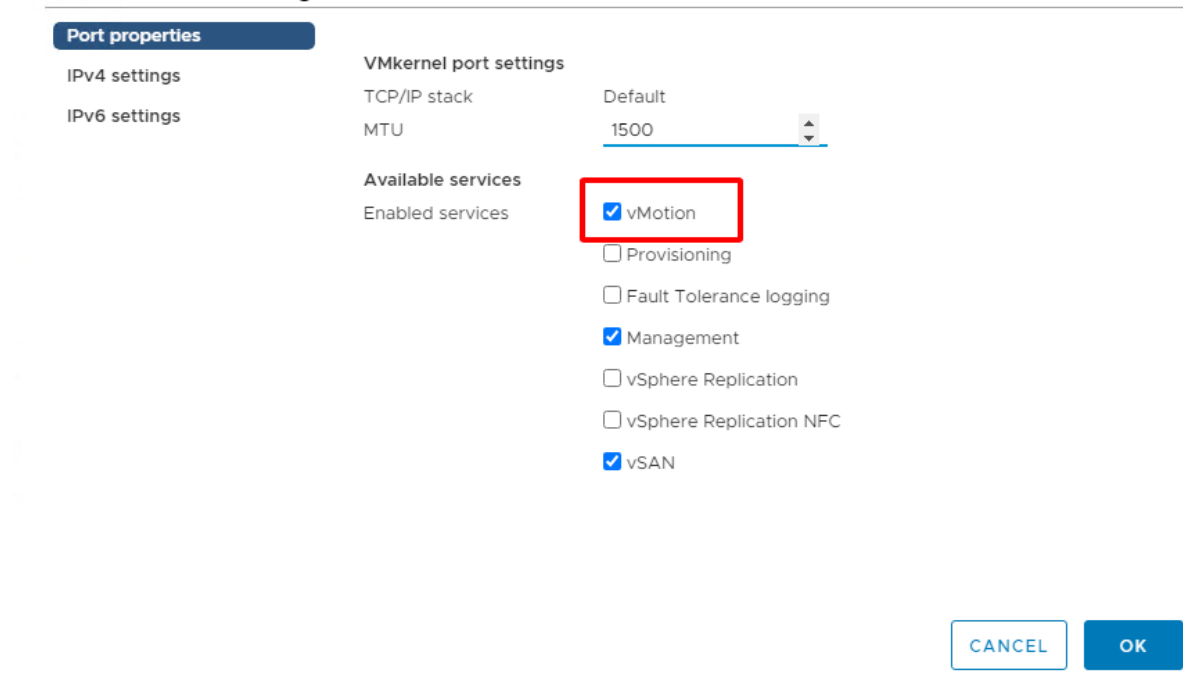
- Thêm thành công chúng ta sẽ thấy 3 hosts



- Sau khi thêm thành công 3 Node vào Cluster ta chọn từng Node bắt chế độ Vmotion lên:



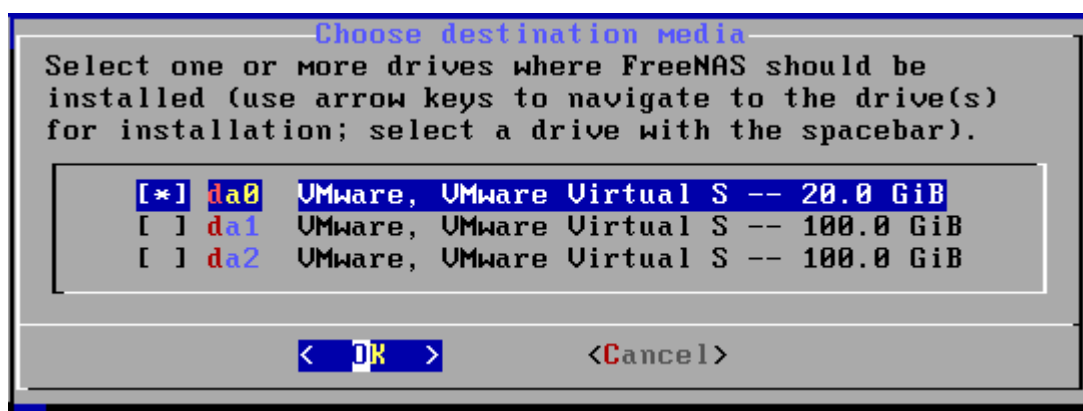
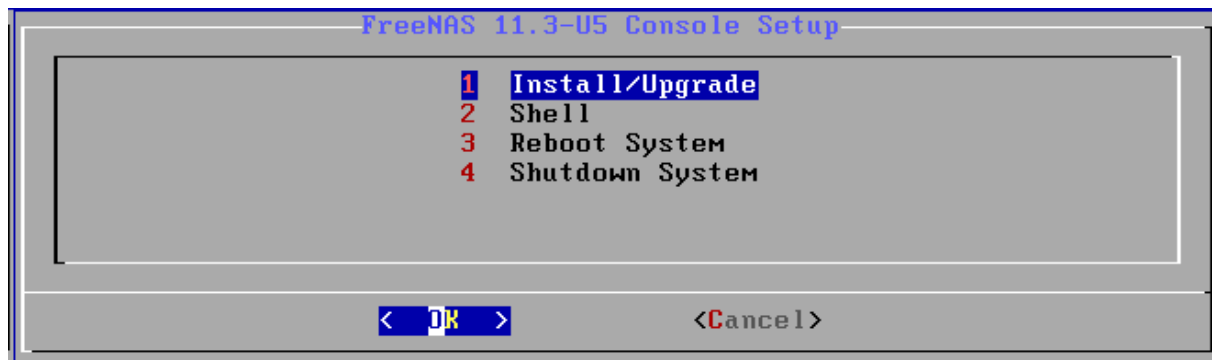
vmk0 - Edit Settings



- Làm tương tự cho các Node còn lại

4. Hướng dẫn Cài đặt và cấu hình FreeNas làm Storage ISCSI

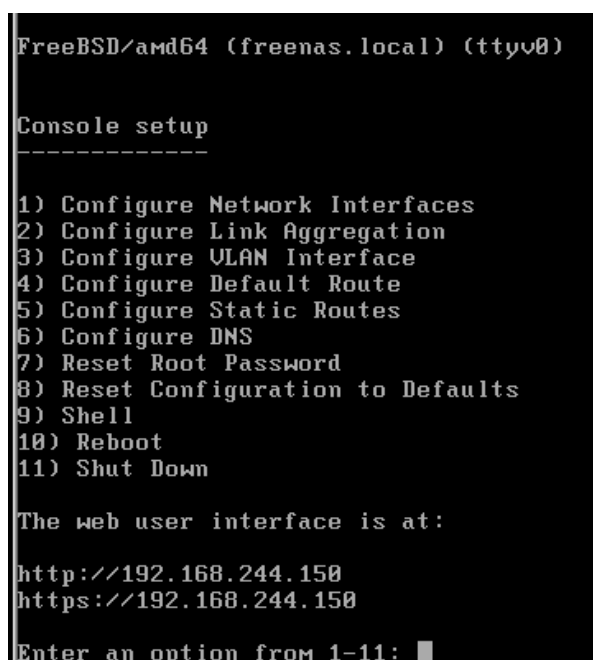
4.1 Cài đặt FreeNas



Chọn ổ cứng để cài hệ điều hành bằng phím cách

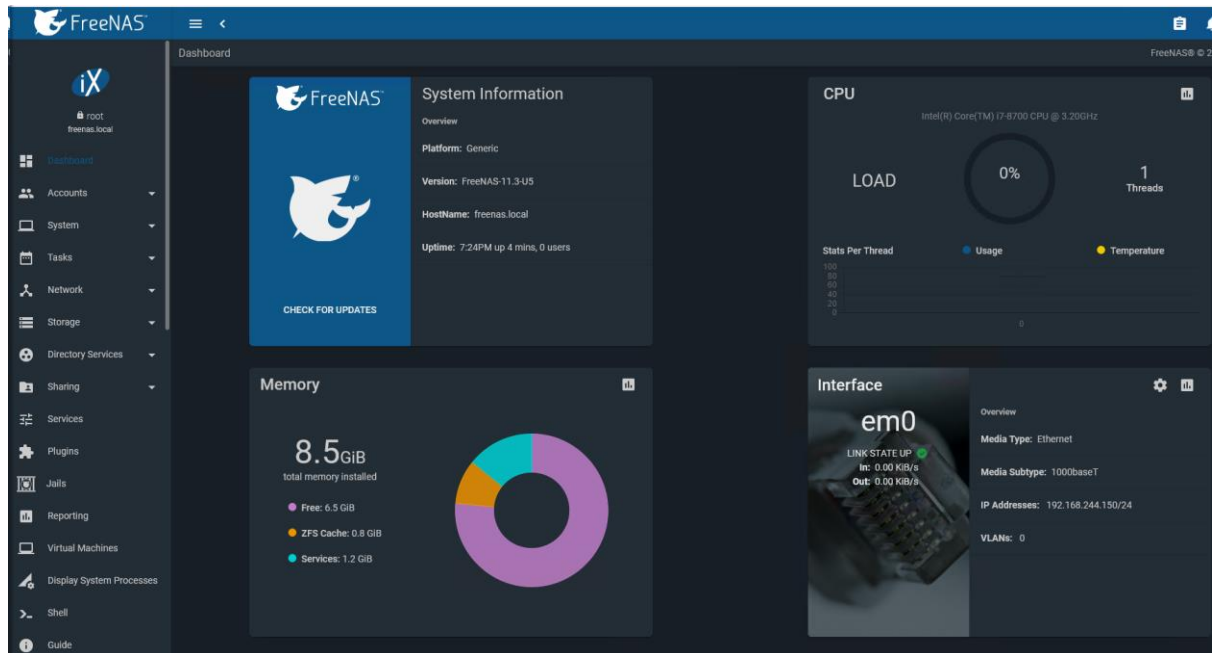
Điền thông tin cần thiết next mặc định cho đến khi finish nhé

Giao diện FreeNas sau khi hoàn thành



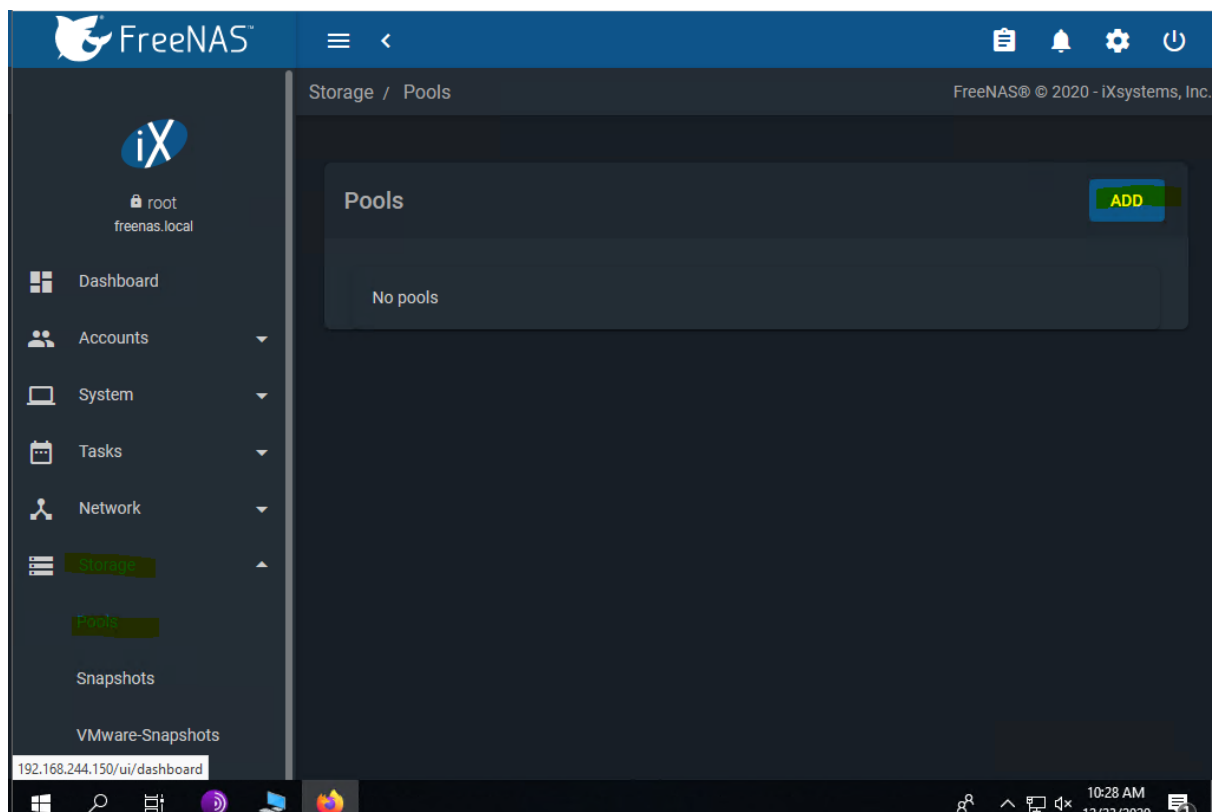
4.2 cấu hình FreeNas

Mở trình duyệt web đăng nhập bằng IP



Các thông số cơ bản như IP, giờ, card mạng các bạn tự mò nhé :D

Vào Storage > Pools > ADD



1 Create or Import... — 2 Decrypt ... — 3 Select pool to i... — 4 Confirm Opt...

Create a pool:

☒ Create new pool ?

☐ Import an existing pool ?

CANCEL **CREATE POOL**

☒	Disk	Type	Capacity	
☒	da1	UNKNO	100 GiB	>
☒	da2	UNKNO	100 GiB	>

2 selected / 2 total

Filter disks by name Filter disks by capacity

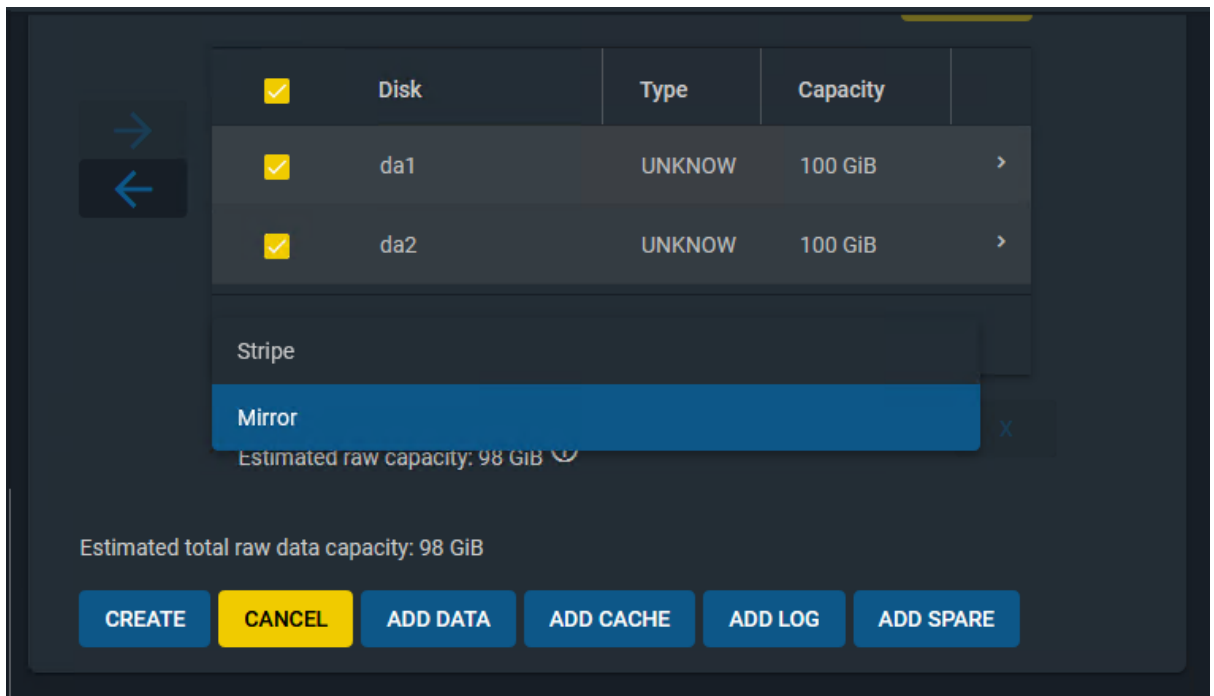
Data VDevs **REPEAT**

☐	Disk	Type	Capacity	
No data to display				
0 selected / 0 total				

→

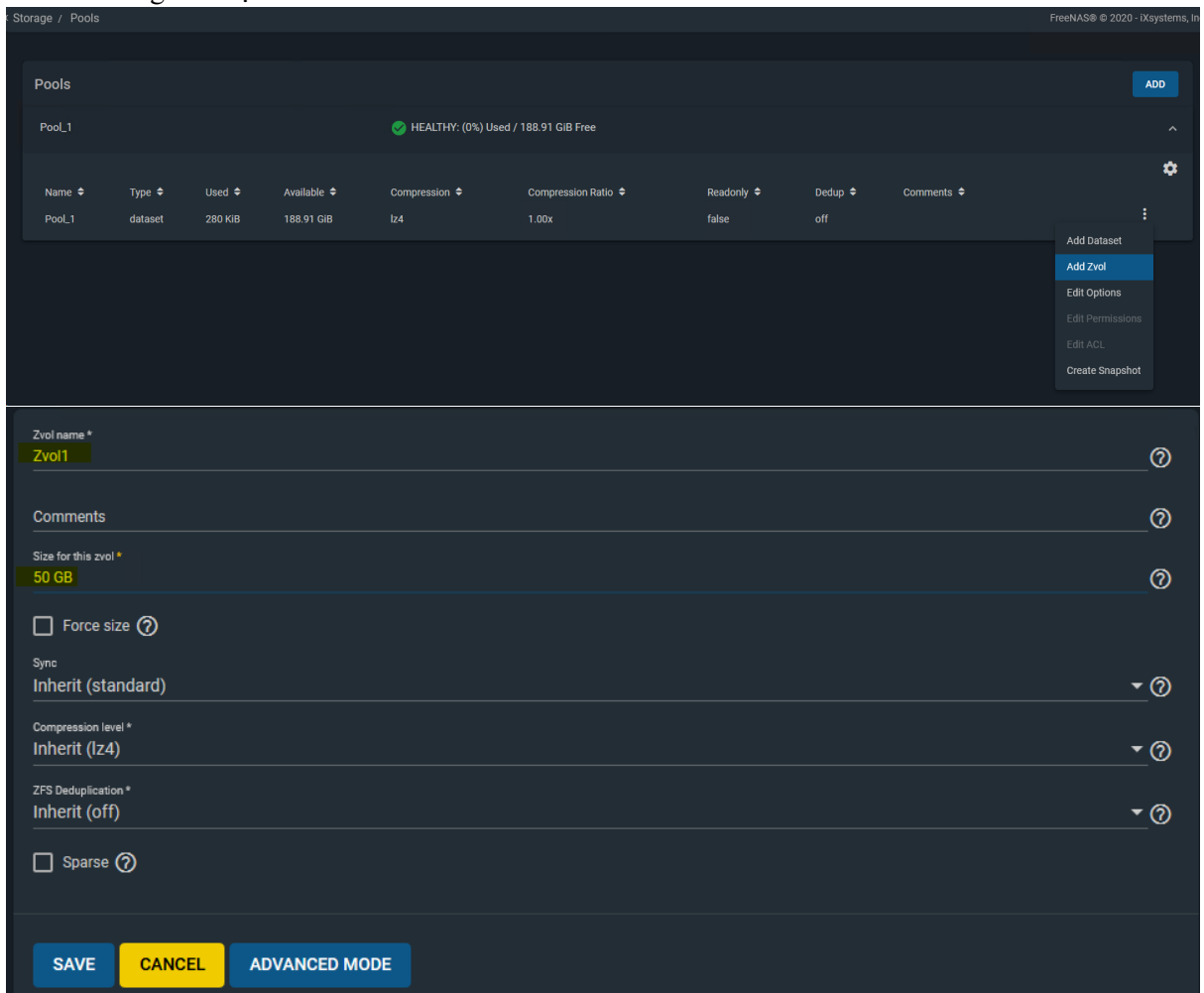
←

Mình có add 2 ổ cứng 100Gb, chọn 2 ổ cứng rồi chọn mũi tên sang phải

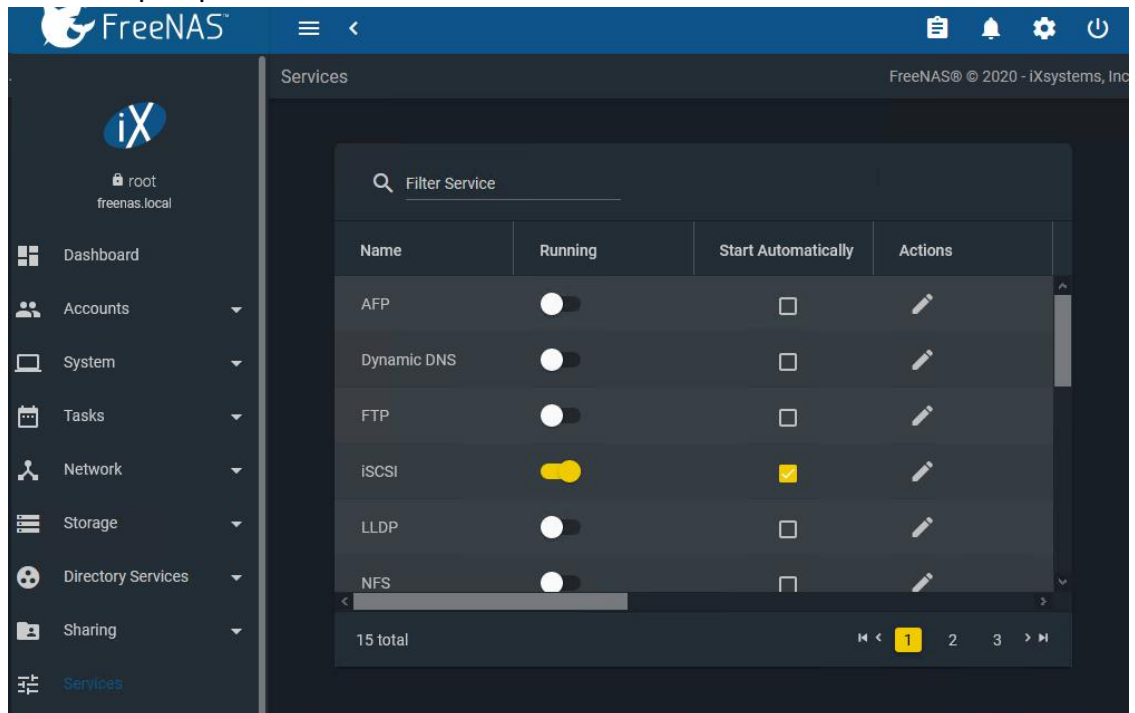


Stripe tương đương raid 0, Mirror tương đương raid 1

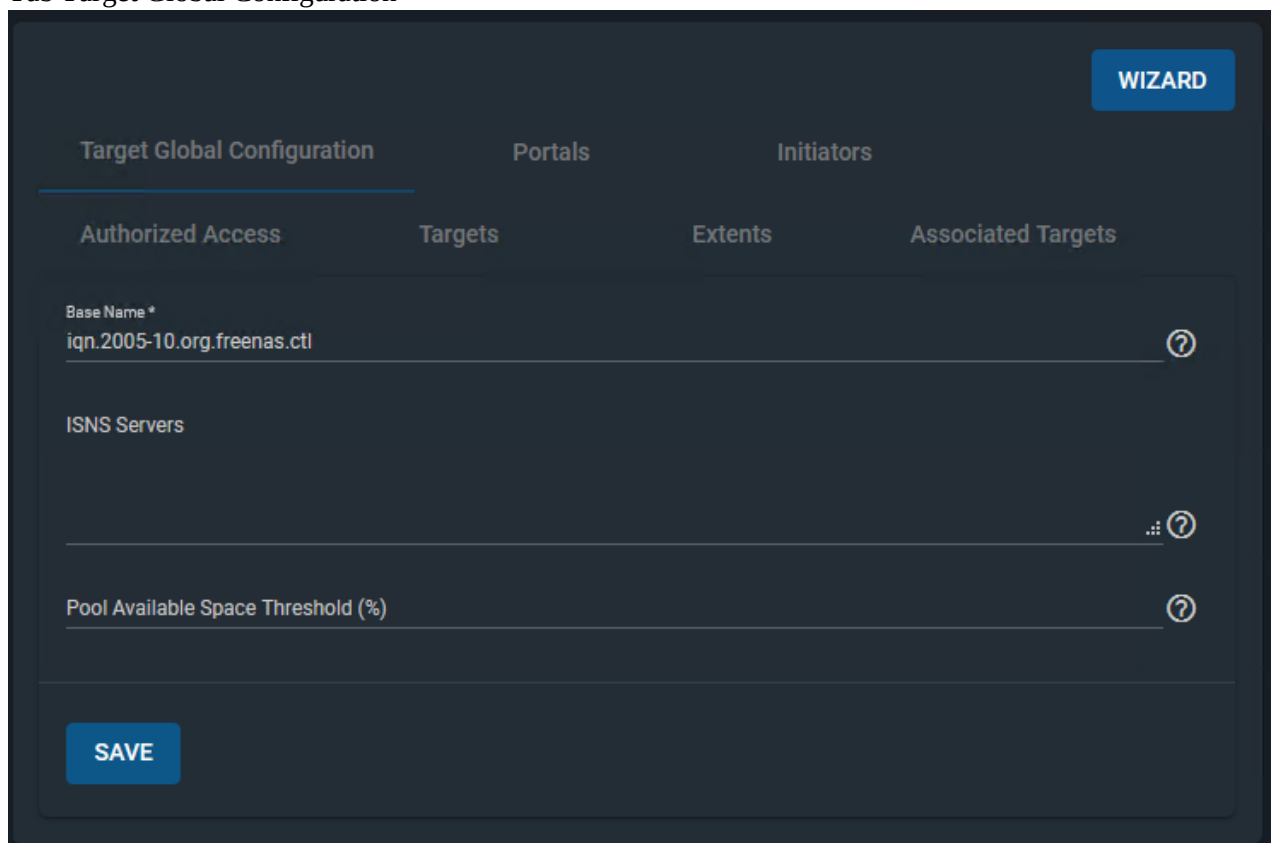
- Sau đó chúng ta sẽ tạo Zvol



- Cấu hình dịch vụ iSCSI



Tab Target Global Configuration



Tab Portals > ADD

Description?

Discovery Auth Method
NONE?

Discovery Auth Group?

IP Address *

0.0.0.0?

Port
3260?

ADD

SAVE

CANCEL

Qua tab Initiators > ADD

☐ Allow All Initiators

Connected Initiators?

→

→

Allowed Initiators (IQN)?

+

ALL

−

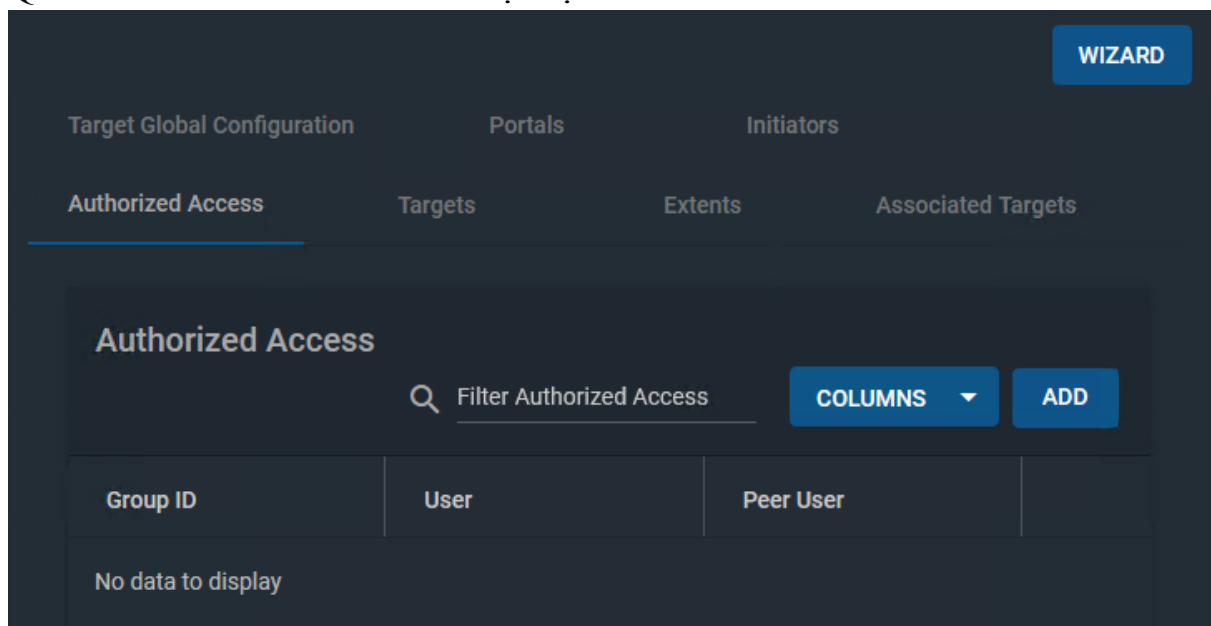
Authorized Networks?

+

192.168.244.0/24

−

Qua tab Authorized Access để mặc định



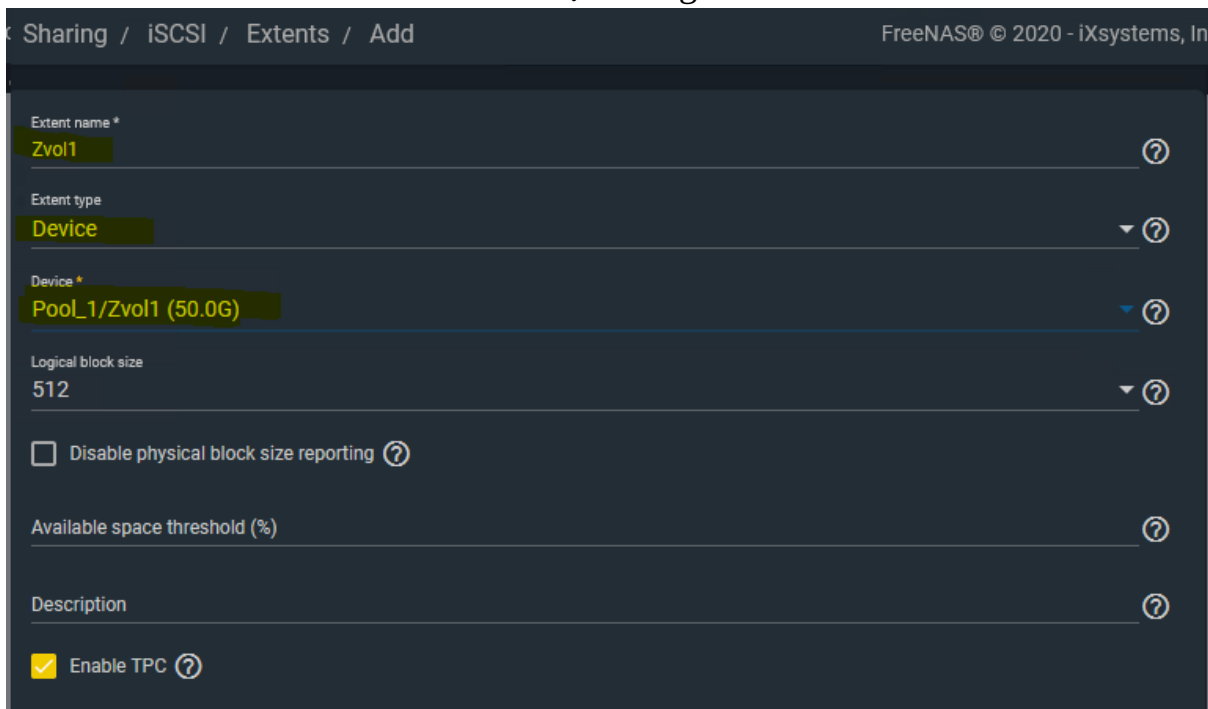
Tab Target > ADD > Điền các thông số như hình

The screenshot shows the "ADD" form for a Target. The form has the following fields:

- Target Name ***: A text input field containing "target1".
- Target Alias**: A text input field.
- Portal Group ID ***: A dropdown menu showing "1".
- Initiator Group ID**: A dropdown menu showing "1 (ALL Initiators Allowed)".
- Auth Method**: A dropdown menu showing "None".
- Authentication Group number**: A dropdown menu.

At the bottom of the form are three buttons: "SAVE" (blue), "CANCEL" (yellow), and "ADD EXTRA ISCSI GROUP" (blue).

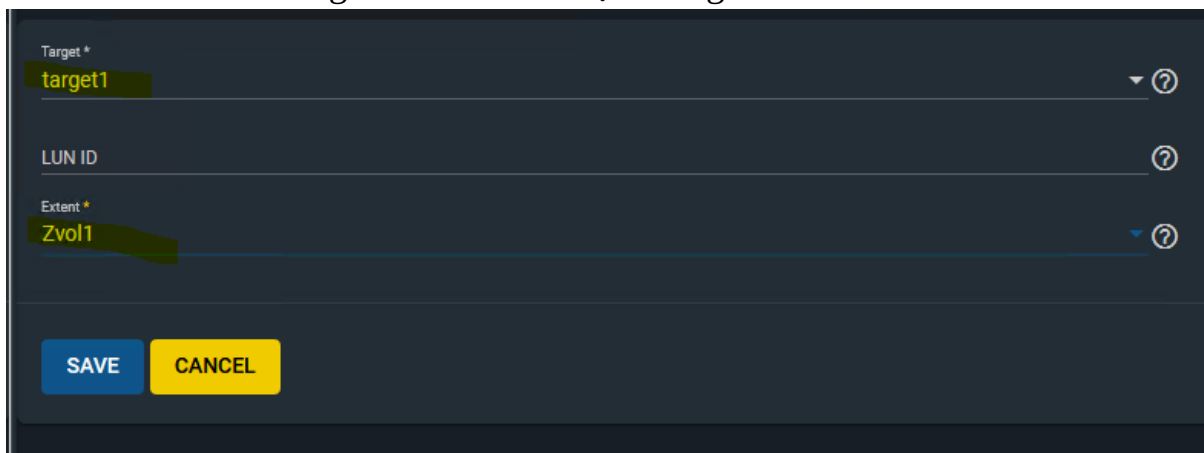
Tab Extents > ADD > Điền tên và chọn thông số như hình



The screenshot shows the 'Add' form for creating a new extent in the FreeNAS web interface. The breadcrumb navigation at the top reads 'Sharing / iSCSI / Extents / Add'. The form fields are as follows:

- Extent name ***: A text input field containing 'Zvol1'.
- Extent type**: A dropdown menu set to 'Device'.
- Device ***: A dropdown menu set to 'Pool_1/Zvol1 (50.0G)'.
- Logical block size**: A dropdown menu set to '512'.
- Disable physical block size reporting**: An unchecked checkbox.
- Available space threshold (%)**: An empty text input field.
- Description**: An empty text input field.
- Enable TPC**: A checked checkbox.

Tab > Associated Targets > ADD > Chọn thông số như hình



The screenshot shows the 'Add' form for creating a new associated target in the FreeNAS web interface. The form fields are as follows:

- Target ***: A dropdown menu set to 'target1'.
- LUN ID**: An empty text input field.
- Extent ***: A dropdown menu set to 'Zvol1'.

At the bottom of the form are two buttons: 'SAVE' (blue) and 'CANCEL' (yellow).

5. Thêm FreeNas vào ESXi

The screenshot shows the vSphere Client interface for host 192.168.244.157. The left sidebar shows the host's configuration tree, with '192.168.244.157 (Maintenance Mode)' selected. The main panel displays the 'Configure' tab, with 'Storage Adapters' highlighted in the left sidebar. The 'Add Software Adapter' button is visible in the top right of the 'Storage Adapters' section.

Add Software Adapter | 192.168.244.157

☒ Add software iSCSI adapter

A new software iSCSI adapter will be added to the list. After it has been added, select the adapter and use the Adapter Details section to complete the configuration.

☐ Add software NVMe over RDMA adapter

Discover software NVMe adapters associated with the following RDMA devices.

☐ Add Software FCoE Adapter

Discover software FCoE adapters associated with the following physical network adapter.

Physical Network Adapter:

VLAN ID: Range: 0 - 4094

Priority Class: Range: 0 - 7

Controller MAC Address:

The screenshot shows the 'Storage Adapters' section for host 192.168.244.157. The 'Add Software Adapter' button is highlighted. Below the table, the 'Dynamic Discovery' tab is selected, and the '+ Add.' button is highlighted.

Adapter	Type	Status	Identifier
Model: iSCSI Software Adapter			
vmhba65	iSCSI	Online	iqn.1998-01.com:vmware:5fe555f2-79a1-904b-3...
Model: PIIX4 for 430TX/440BX/MX IDE Controller			
vmhba1	Block SCSI	Unknown	--
vmhba64	Block SCSI	Unknown	--

Properties | Devices | Paths | **Dynamic Discovery** | Static Discovery | Network Port Binding | Advanced Optic

+ Add. | X Remove | Authentication... | Advanced...

iSCSI server

- Nhập ip của FreeNas

Add Send Target Server
vmhba65

iSCSI Server:
192.168.244.142

Port:
3260

☒ Inherit authentication settings from parent

- Qua tab Static Discovery thấy xuất hiện thiết bị mới là ok

Storage Adapters

Due to recent configuration changes, a rescan of "vmhba65" is recommended.

+ Add Software Adapter
Refresh
Rescan Storage...
Rescan Adapter
Remove

Adapter	Type	Status	Identifier	Targets
Model: iSCSI Software Adapter				
vmhba65	iSCSI	Online	iqn.1998-01.com.vmware:5fe555f2-79a1-904b-3...	1
Model: PIIX4 for 430TX/440BX/MX IDE Controller				

Properties
Devices
Paths
Dynamic Discovery
Static Discovery
Network Port Binding
Advanced Options

+ Add...
Remove
Authentication...
Advanced...

iSCSI server	Target Name
192.168.244.142:3260	iqn.2005-10.org.freenas.cti:tag

- Trở lại Storage Adapters > Rescan Storage...

Rescan Storage
192.168.244.157

☒ Scan for new Storage Devices

Rescan all host bus adapters for new storage devices. Rescanning all adapters can be slow.

☒ Scan for new VMFS Volumes

Rescan all known storage devices for new VMFS volumes that have been added since the last scan. Rescanning known storage for new file systems is faster than rescanning for new storage.

CANCEL
OK

- Click phải vào node 1 (244.157) > Storage > New Datastore

New Datastore

1 Type

2 Name and device selection

3 VMFS version

4 Partition configuration

5 Ready to complete

Type

Specify datastore type.

☒ VMFS

Create a VMFS datastore on a disk/LUN.

☐ NFS

Create an NFS datastore on an NFS share over the network.

☐ vVol

Create a Virtual Volumes datastore on a storage container connected to a storage provider.

New Datastore

✓ 1 Type

2 Name and device selection

3 VMFS version

4 Partition configuration

5 Ready to complete

Name and device selection

Select a name and a disk/LUN for provisioning the datastore.

Datastore name: **Datastore_SAN**

Name	LUN	Capacity	Hardware...	Drive T...	S
Local VMware, Disk (mpx...	0	200.00 GB	Not suppor...	HDD	5
FreeNAS ISCSI Disk (naa...	0	220.00 GB	Supported	Flash	-
Local VMware, Disk (mpx...	0	100.00 GB	Not suppor...	HDD	5

New Datastore

✓ 1 Type

✓ 2 Name and device selection

3 VMFS version

4 Partition configuration

5 Ready to complete

VMFS version

Specify the VMFS version for the datastore.

☒ VMFS 6

VMFS 6 enables advanced format (512e) and automatic space reclamation support.

☐ VMFS 5

VMFS 5 enables 2+TB LUN support.

New Datastore

✓ 1 Type

✓ 2 Name and device selection

✓ 3 VMFS version

4 Partition configuration

5 Ready to complete

Partition configuration

Review the disk layout and specify partition configuration details.

Partition Configuration

Use all available partitions

Datastore Size

220

GB

Block size

1 MB

Space Reclamation Granularity

1 MB

Space Reclamation Priority

Low: Deleted or unmapped blocks are reclaimed on the LUN at Low priority

HPFS/NTFS: 219.5 GB

HPFS/NTFS

HPFS/NTFS

This configuration will delete the current disk layout, change the file system and change the partition table format. All file systems and data will be permanently lost.

CANCEL

BACK

NEXT

New Datastore

✓ 1 Type

✓ 2 Name and device selection

✓ 3 VMFS version

✓ 4 Partition configuration

5 Ready to complete

Ready to complete

Review your settings selections before finishing the wizard.

General

Name:

Datastore_SAN

Type:

VMFS

Datastore size:

220.00 GB

Device and Formatting

Disk/LUN:

FreeNAS iSCSI Disk
(naa.6589cfc000000bbdeea50e17adccdc10)

Partition Format:

GPT

VMFS Version:

VMFS 6

Block Size:

1 MB

Space Reclamation Granularity:

1 MB

Space Reclamation Priority

Low: Deleted or unmapped blocks are reclaimed on the LUN at low priority

- Làm đúng sẽ xuất hiện Datastore_SAN của FreeNas

The screenshot shows the vSphere Datastore browser interface. On the left, a tree view shows the hierarchy: 192.168.244.100 > Datacenter > Cluster 1 > 192.168.244.157 (Maintenance Mode). The 'Datastores' tab is selected at the top. The main table lists two datastores:

Name	Status	Type	Capacity	Free
Datastore_SAN	✓ Normal	VMFS 6	219.75 GB	218.34 GB
vsanDatastore	✓ Normal	vSAN	0 B	0 B

- Làm tương tự cho những node muốn thêm vào cluster.
- Giờ mình sẽ tiến hành cài VM lên storage SAN (cài lên node 244.158)
- Tới bước chọn nơi lưu trữ nhớ chọn là Datastore_SAN nhé

New Virtual Machine

The screenshot shows the 'New Virtual Machine' wizard. On the left, a list of steps is shown, with '4 Select storage' highlighted. The main area is titled 'Select storage' and includes a checkbox for 'Encrypt this virtual machine (Requires Key Management Server)'. Below this, the 'VM Storage Policy' section shows a table of available storage policies:

Name	Capacity	Provisioned	Free	Type
Datastore_SAN	219.75 GB	1.41 GB	218.34 GB	VMFS 6
vsanDatastore	0 B	0 B	0 B	vSAN

- Test: VM_SAN đã được cài lên Node 244.158 giờ ta sẽ tiến hành tắt Node 244.158 sau khoảng hơn 2 phút thì VM_SAN sẽ tự chuyển sang 1 trong 2 Node còn lại là thành công.